

BlackRock industry project • UC Berkeley Haas MFE

Macroeconomic Surprises, DTW Price Patterns and the Intraday Response of ES, NQ, YM and ZN Futures

E-mini S&P 500 (ES) • E-mini Nasdaq-100 (NQ) • E-mini Dow (YM) • 10-Year Treasury Note (ZN)

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Structure of the presentation

1

The answer and the questions

Headline results · motivation · 7 research questions

slides 3–4

2

Markets, signals and method

ES, NQ, YM, ZN · jump vs drift · surprise, DTW, state · walk-forward design

slides 5–9

3

RQ1–RQ2 · What moves, what is tradeable

Notebooks 02, 02b: CPI dominates; the drift after the first minute

slides 10–12

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RQ3 · Pre-event market state

Notebooks 03, 03b: volatility and the CPI response

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RQ4 · CPI walk-forward strategy

Notebooks 04, 04b: performance, regime, real-time detection, robustness

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RQ5 · Scaling to ~10 releases

Notebooks 05, 05b: ~100 trades a year per market

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RQ7 · Portfolio and team comparison

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Discussion and conclusions

Why equities not bonds · final comparison · recommendations · future work

slides 36–39

Slide 40 · Acknowledgements and questions

The answer in one slide

Walk-forward 2019–2026, entry one minute after the release, net of 2 ticks + \$4.50 per round trip

≈ 1.0

Net Sharpe of the CPI surprise strategy on ES (0.8–1.0 on NQ and YM)

0.78

Net Sharpe of the ES + NQ + YM multi-release book, ~290 trades a year

30 min

How long the tradeable drift lasts after the first minute

0 / 0

Out-of-sample gain from pre-event state / from DTW patterns

Works

- Macro surprise in equity-index futures (ES, NQ, YM)
- CPI is the core release
- ~100 trades a year per market with ~10 releases

Does not help

- Pre-event state: predicts the size of the move, not its direction
- DTW price patterns: no reliable directional information (YM blend 1.06, not significant)
- More windows per release

Does not work

ZN (10-year T-note): priced in the first minute; every strategy loses after costs

Why macro releases, and what we asked

Scheduled information shocks: known time, published expectation, immediate repricing

The opportunity and the problem

- The release time and the consensus are known in advance
- Theory predicts the direction: hot inflation → higher rates → lower equities and bond prices
- But the first move happens in milliseconds: a strategy that decides after seeing the number can only trade what comes after the first minute

RQ1

Which releases move ES, NQ, YM and ZN, and by how much?

RQ2

How much of the reaction is tradeable after the first minute?

RQ3

Does the pre-event state change the reaction?

RQ4

Can it be traded in real time after costs? Does state help?

RQ5

How many trades a year across many releases?

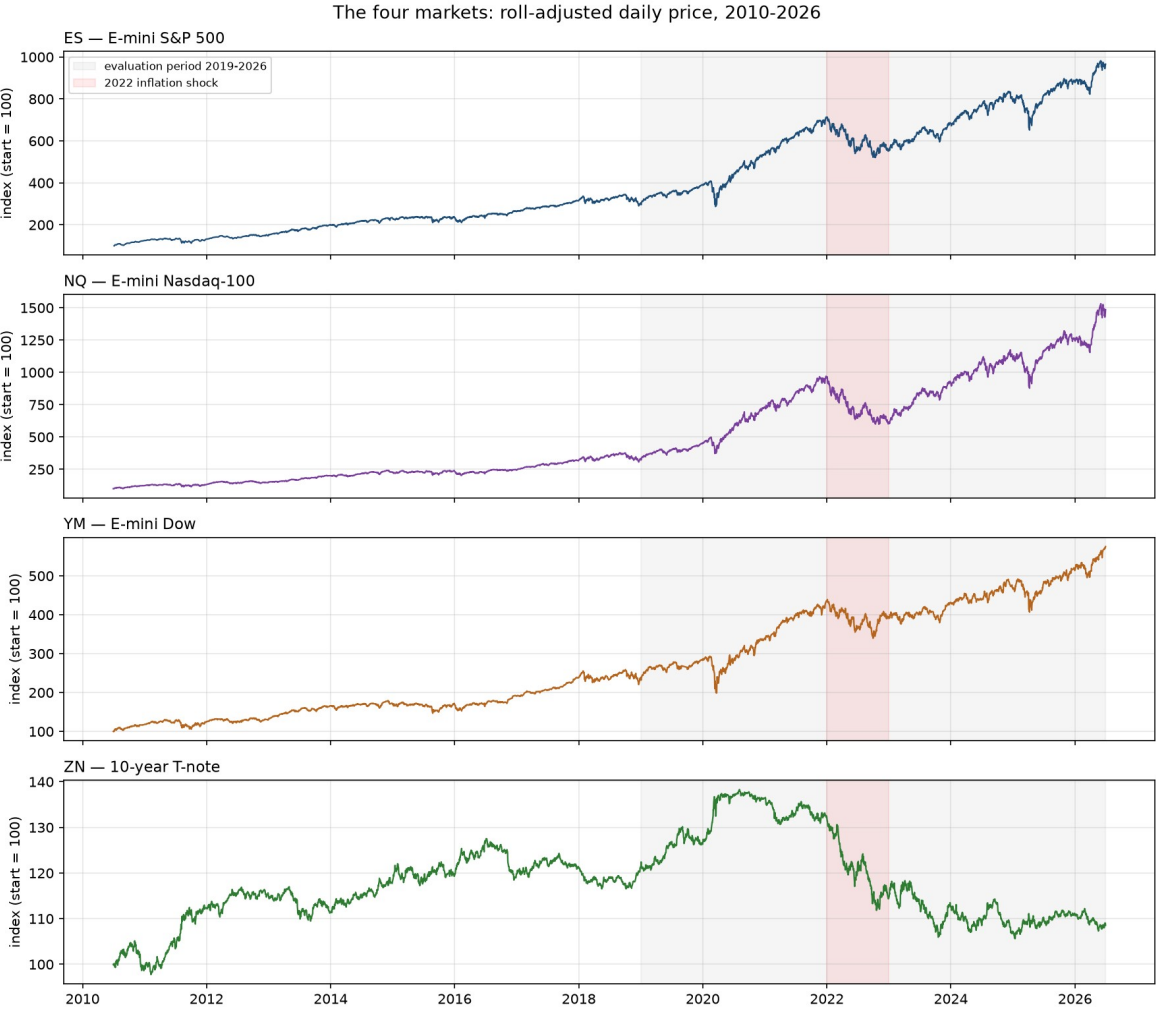
RQ6

Does DTW add information? Do more windows help?

RQ7

Do markets combine? How does the team compare at equal costs?

Four markets, three ways to react to the same news



	ES	NQ	YM	ZN
Underlying	S&P 500	Nasdaq-100	Dow	10-yr T-note
Value of a tick	\$12.50	\$5.00	\$5.00	\$15.63
Cost (2 ticks + \$4.50)	≈1.7 bps	≈0.7 bps	≈1.0 bps	≈3.0 bps
Buy & hold 2019–26	+218%	+348%	+141%	−10.5%
Hot CPI print	down	down more	down	down (yields up)

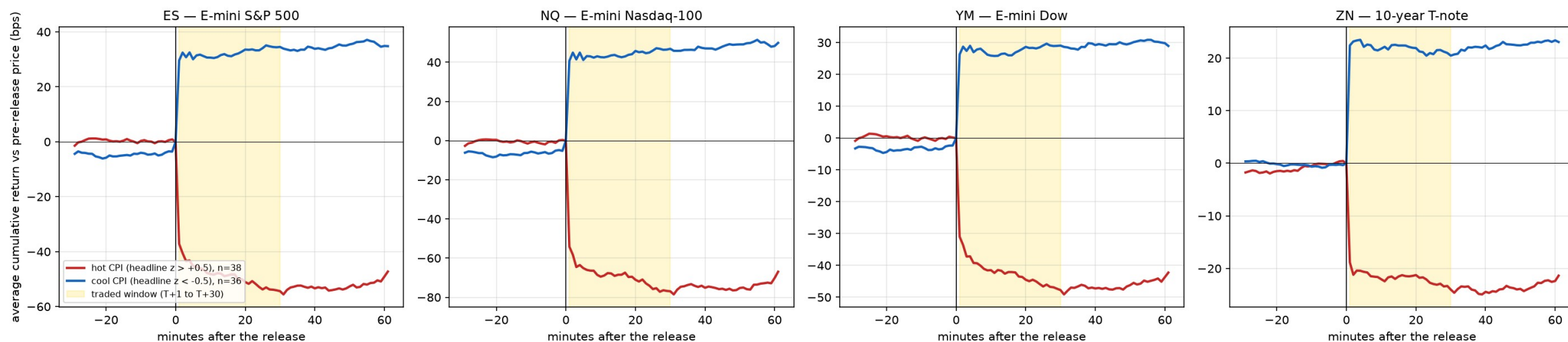
Why these four

- Is the effect a property of one contract, of equity indices, or of rate-sensitive markets?
- NQ: long-duration tech → more rate-sensitive
- YM: the same index family with more industrials and banks, added later as an out-of-sample check

A release produces a jump, then a drift

Average path around 74 CPI releases with $|\text{surprise}| > 0.5 \text{ SD}$, 2016–2026

Average market path around CPI releases (2016–2026): a jump at the release, then a smaller drift



-37 → -55 bps

ES after a hot CPI print: first-minute jump, then T+30

-55 → -77 bps

NQ: larger jump and larger drift

-31 → -48 bps

YM: a slightly smaller ES

-19 → -24 bps

ZN: almost nothing left after the first minute

Two signal families, plus the market state

1 • Macro surprise

- Surprise = actual – Bloomberg consensus
- Standardised causally by the SD of previous surprises (≥ 24), capped at $\pm 5\sigma$
- Line signs from the historical jump; lines released together form a family
- $x > 0$ always means good news for that market (NFP sign reversed for ZN)

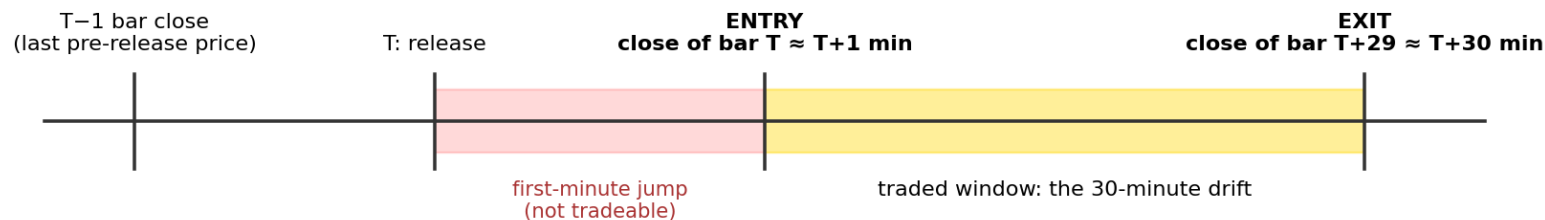
2 • DTW price pattern (team spec)

- Query: z-scored 31-minute path ending at entry
- Library: earlier releases at the same clock time, 3-year lookback, ≤ 600
- $k = 15$ nearest, band 5, weights $1/\text{distance} \times 2\text{-year half-life}$
- Trade the neighbours' average next-30-min return if outside the middle 40%

3 • Pre-event state

- Six pre-registered states: 20-day vol, 1-day vol, 20-day momentum, 60-day drawdown, overnight move, previous surprise
- Causal percentiles against a trailing one-year window
- Used to scale or filter the surprise trade

Method: executable, walk-forward, cost-aware



Entry

Close of the first post-release bar ($\approx$ T+1 min); exit at T+30. The jump is never traded

Walk-forward

Every sign, screen, model and gate uses only earlier releases; evaluation 2019–2026

Costs

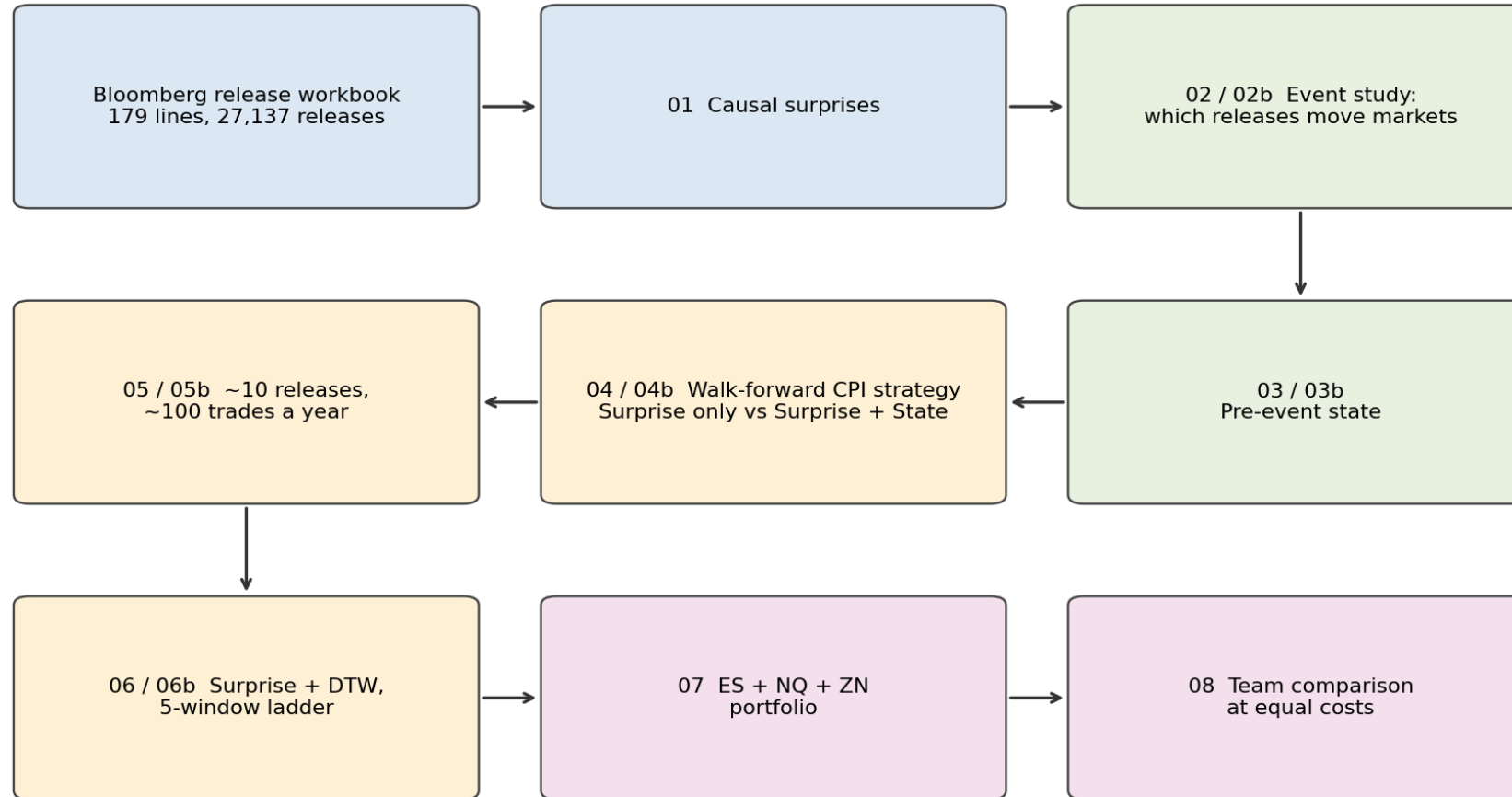
2 ticks + \$4.50 per round trip; sensitivity 0–8 ticks and the team's $\frac{1}{4}$ tick per side

Statistics

HC3 errors, BH-FDR, permutation and random-sign tests, block-bootstrap Sharpe

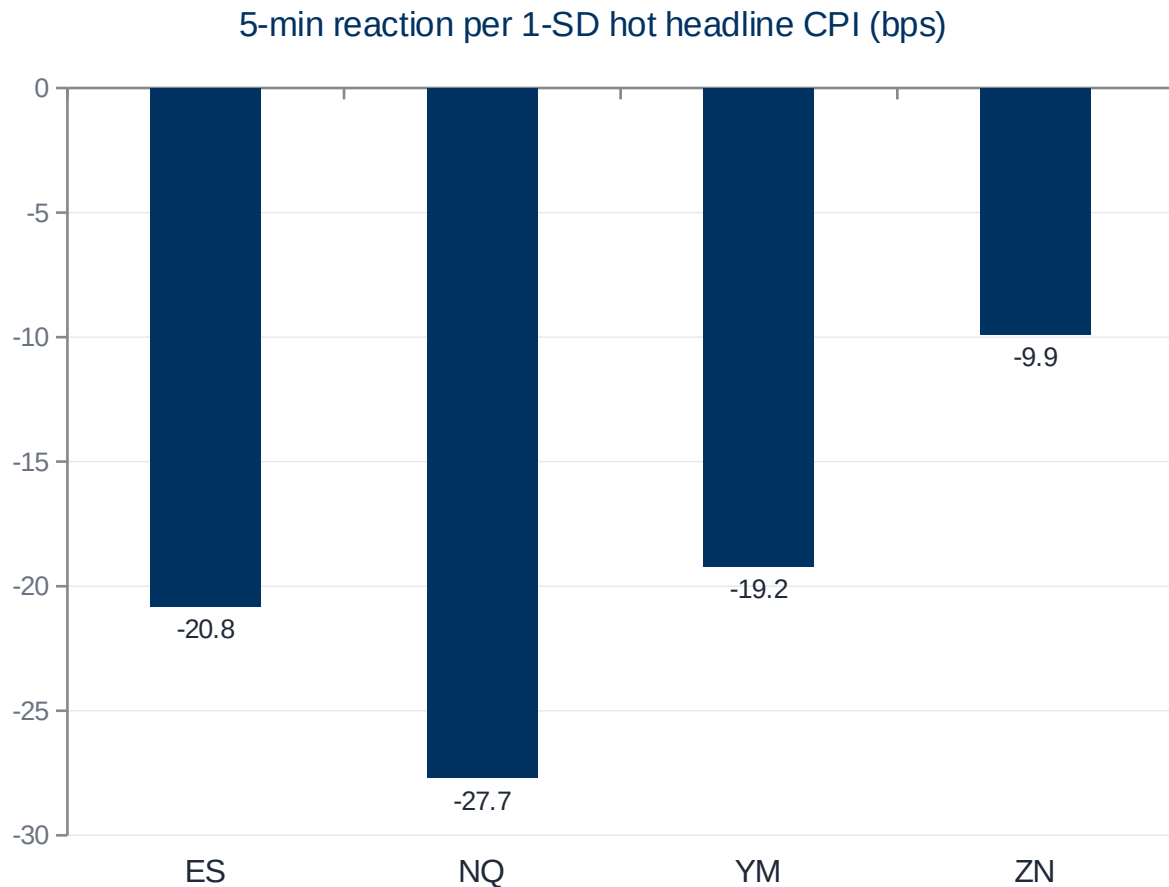
Research pipeline: 8 notebooks, repeated for NQ, YM and ZN

Market data: ES / NQ / ZN 1-minute bars → roll-adjusted log prices (used from Notebook 02 onwards)



RQ1 · CPI is the release that moves all four markets

Notebooks 02 and 02b: 363 CPI, NFP and PCE releases, 2016–2026



	CPI	NFP	PCE
Mean 30-min move , ES	38.5 bps	29.6 bps	14.7 bps
R ² , 5 min, ES	0.27	0.03	0.10
R ² , 5 min, NQ	0.30	0.01	0.10
R ² , 5 min, YM	0.27	0.07	0.09
R ² , 5 min, ZN	0.30	0.03	0.04

Takeaway

- CPI: large move and a predictable direction in every market
- NFP moves markets as much, but its surprise explains only 1–3%: jobs, unemployment and wages often disagree
- NQ reacts ~35% more than ES (rate-sensitive tech); YM ~92% of ES

RQ1 · How much ES moves, and how much the surprise explains

Notebook 02, ES: 122 CPI, 122 NFP and 121 PCE releases, 2016–2026

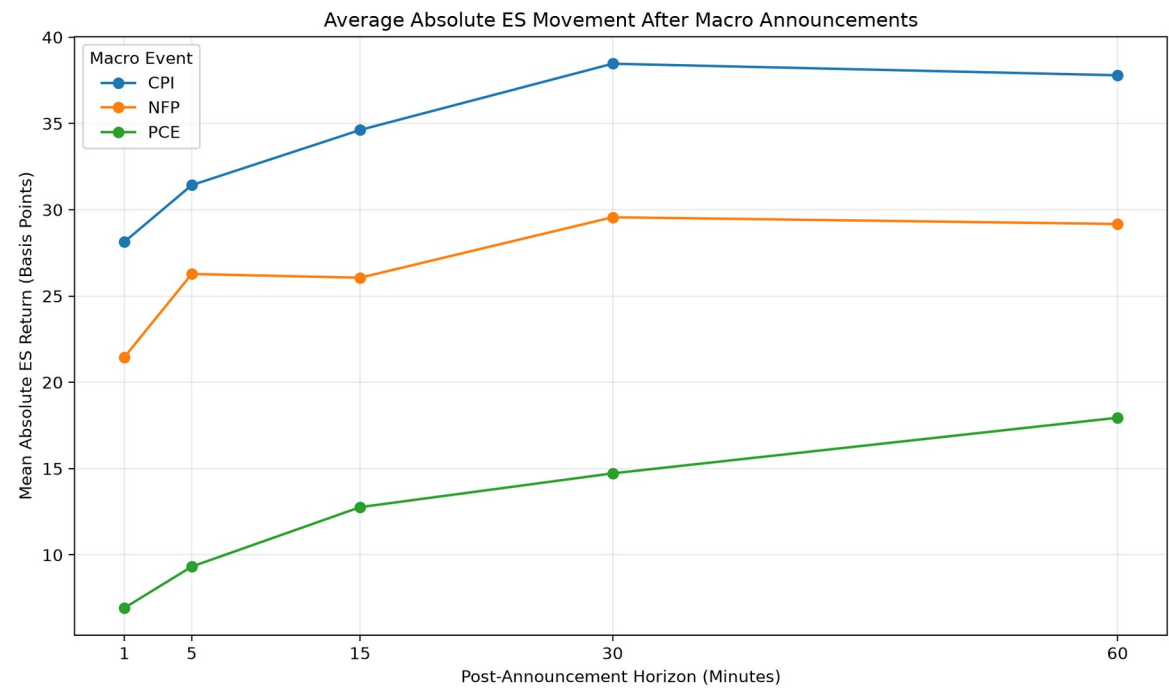


Figure 3 · Mean absolute ES reaction by release and horizon (bps)

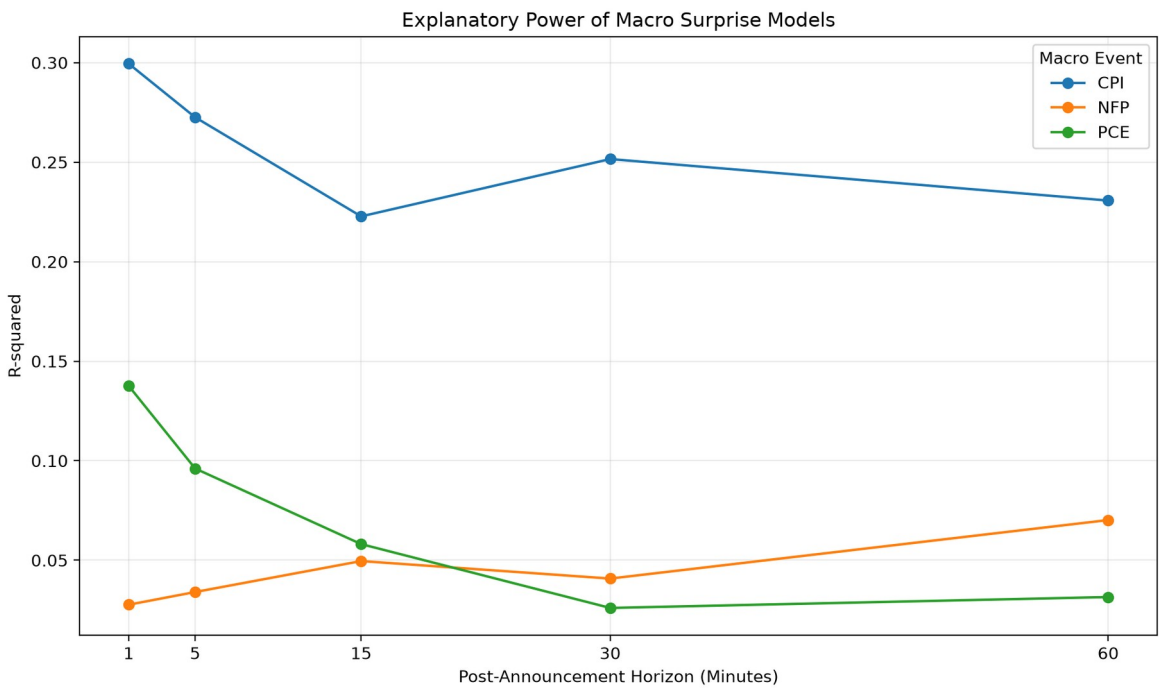
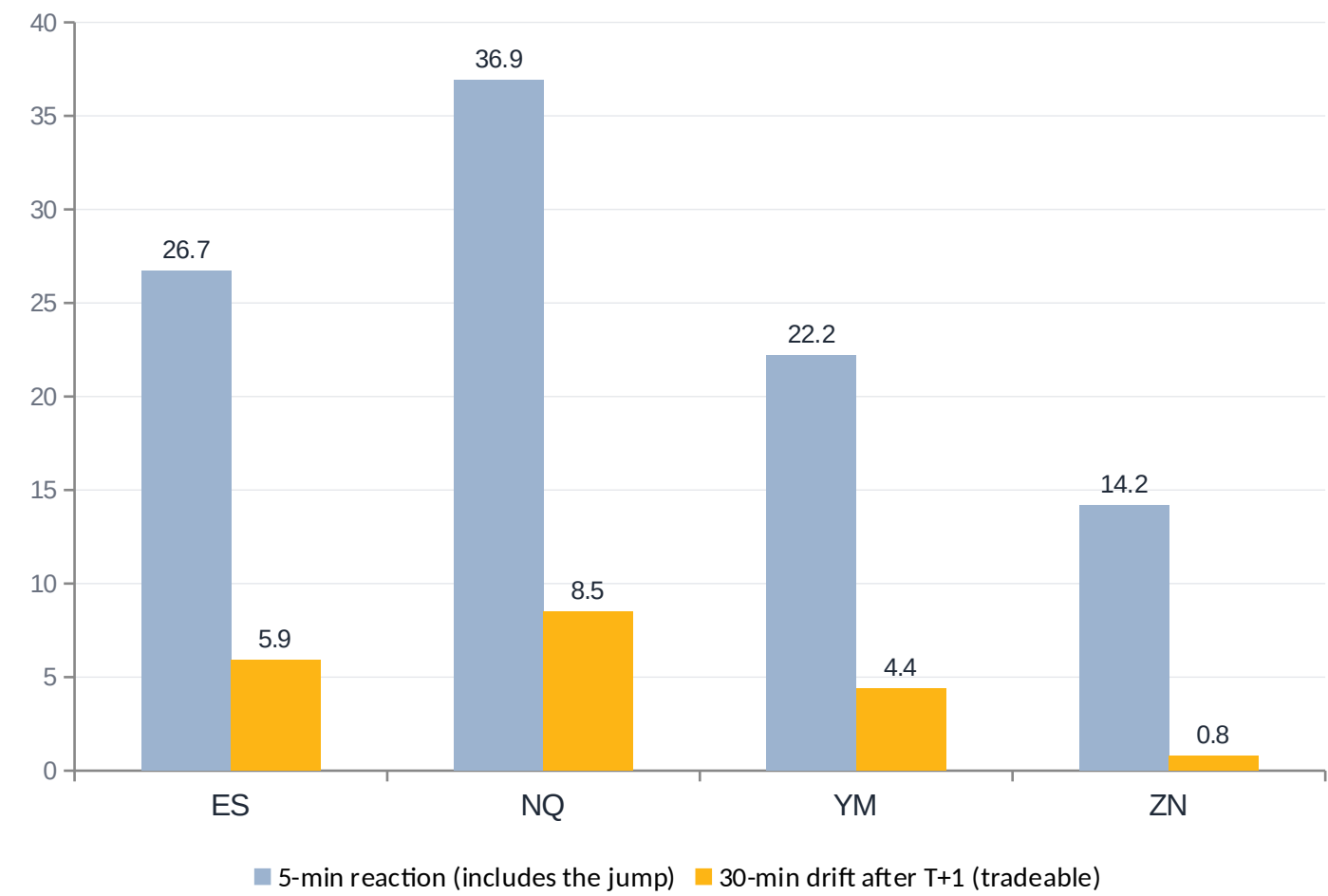


Figure 4 · Share of the ES reaction explained by the surprise (R^2)

CPI moves ES most (38 bps at 30 min) and its surprise explains 22–30% of the move at every horizon. NFP moves ES almost as much (30 bps) but explains only 3–7%; PCE is informative only in the first minutes.

RQ2 · Most of the reaction is untradeable

CPI surprise beta (bps per 1 SD): 5-minute reaction from the pre-release price vs 30-minute drift after entry



$\sim \frac{1}{5} - \frac{1}{4}$

of the CPI reaction remains as a drift in ES, NQ and YM

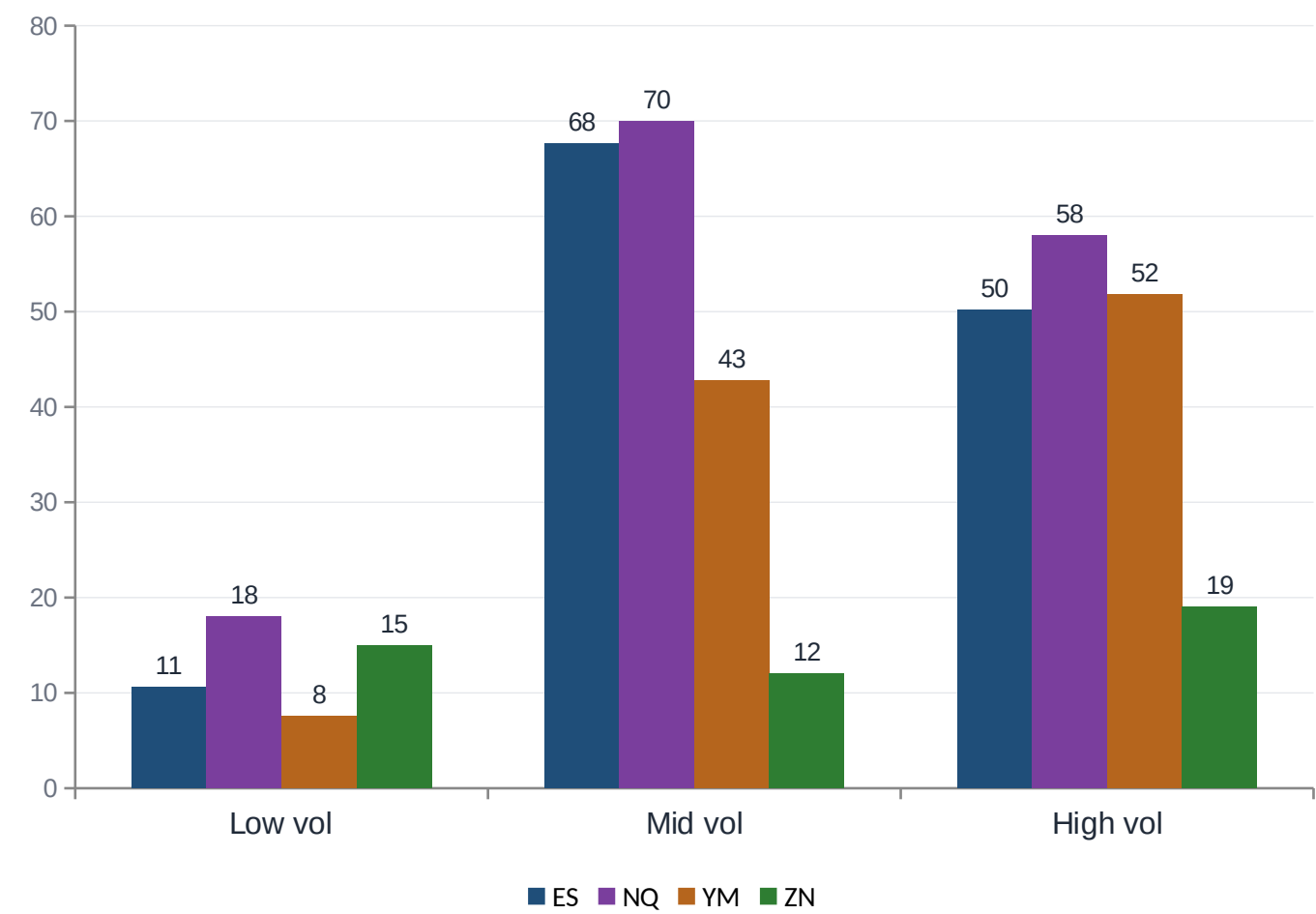
0.8 bps

ZN drift per 1 SD, below its 3 bps cost

Drift vs cost decides the rest of the study: ES 5.9 vs 1.7 bps, NQ 8.5 vs 0.7 bps, YM 4.4 vs 1.0 bps, ZN 0.8 vs 3.0 bps.

RQ3 · In volatile markets CPI hits equities much harder

Notebook 03: CPI surprise beta on the 5-min reaction by tercile of 20-day volatility (bps per 1 SD)

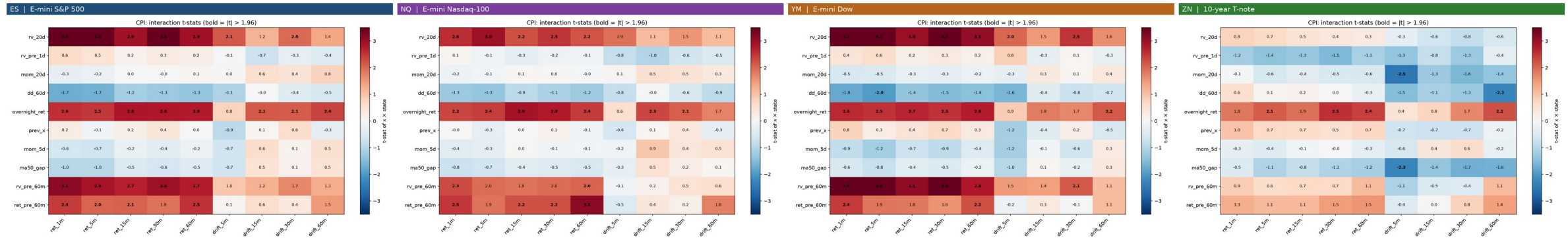


	ES	NQ	YM	ZN
Best interaction t	3.8	3.0	3.7	-3.2
Survive FDR	2	0	3	1
Size tests surviving FDR	198/360	154/360	156/360	0/360

But

- The effect is mostly on the first-minute jump
- Significant only after 2021
- The state reliably predicts the size of the move, not its direction

RQ3 · The same pattern in ES, NQ and YM, a different one in ZN

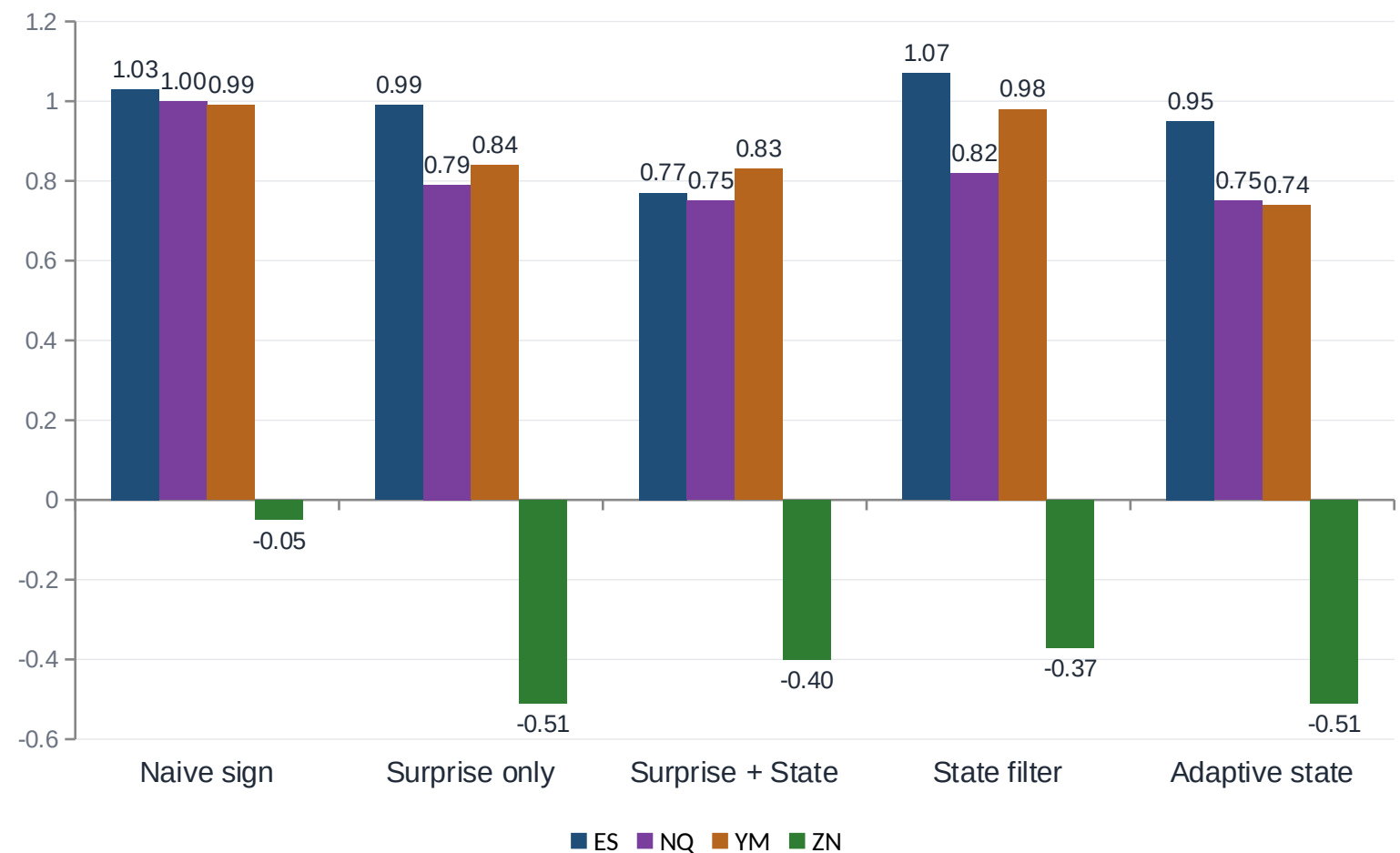


Reading the t-statistics (CPI × state; rows = states, columns = horizons)

- ES: 20-day volatility row dark red on the reactions (t up to 3.8), much lighter on the drifts
- NQ: the same red rows, slightly weaker (t up to 3.0) — not a fluke of one contract
- Overnight return strengthens the response in all three markets
- ZN: no volatility effect; strong recent trends weaken the post-release drift
- Selected for trading: ES $rv_{20d} \times CPI$; NQ none; ZN 1-day vol × all releases

RQ4 · CPI surprises are tradeable in ES, NQ and YM, not in ZN

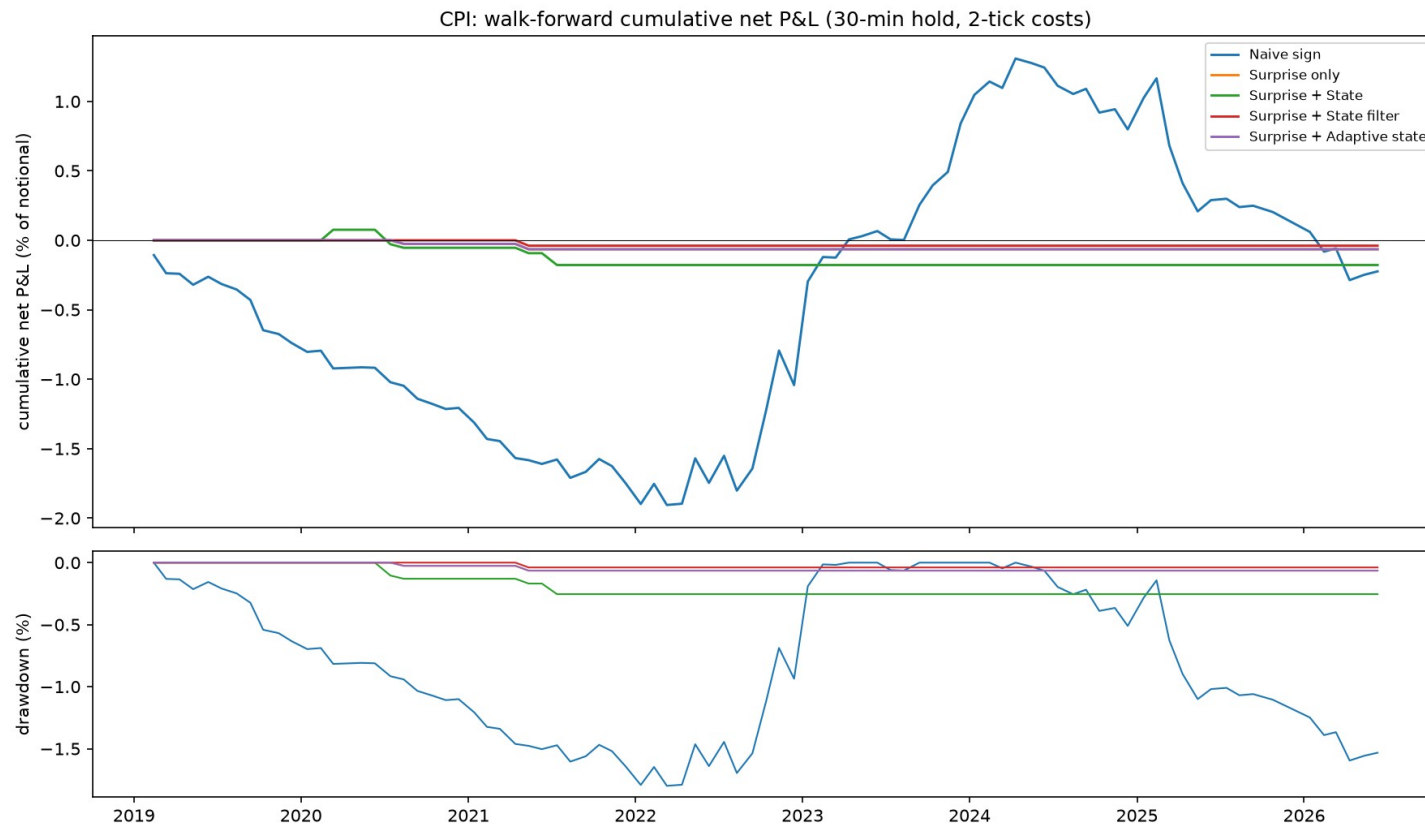
Notebook 04: CPI walk-forward net Sharpe, 2019–2026, 2 ticks + \$4.50



Takeaways

- ES surprise only: 12.8 bps per trade, t = 2.8, CI 0.25–1.72
- The naive sign is as good as any fitted model (NQ: significantly better; YM 0.99)
- No state model beats surprise only
- ZN: +2.7 bps gross per trade, eaten by the 3 bps cost

RQ4 · ZN: the fitted models correctly refuse to trade



+2.7 bps

Naive CPI gross profit per trade in ZN (gross Sharpe 0.53)

-0.3 bps

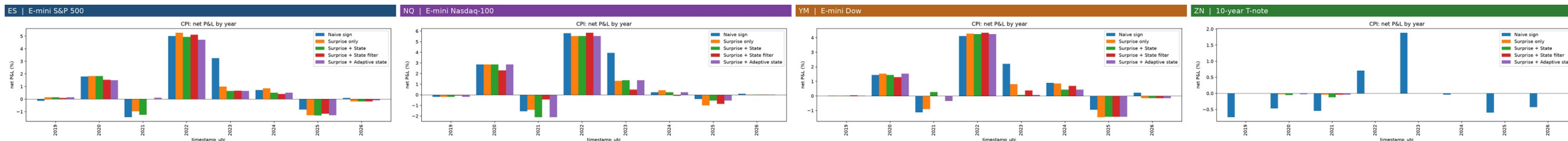
the same after the 3 bps round-trip cost (Sharpe -0.05)

Why the lines are flat

- The surprise model trades only if the predicted move exceeds the cost
- In ZN that happened 1–5 times in 7.5 years

Figure 16 · ZN: walk-forward cumulative net P&L of the CPI strategies and drawdowns

RQ4 · The edge is concentrated in the 2022 inflation shock



CPI net P&L by year (% of notional), five strategies: ES, NQ, YM, ZN

79% / 74% / 86%

Share of CPI P&L from 2022 (ES / NQ / YM, surprise only)

0.58 / 0.68 / 0.53

Naive CPI Sharpe excluding 2022 (ES / NQ / YM)

Why it matters

- CPI pays when inflation drives policy expectations
- The state effect was invisible in real time until 2022 ($t = 0.5$ in 2019, 3.3 in 2022)
- The 'best' state changed every few years: none → 20-day vol → overnight move
- Implication: scale exposure with the macro regime, not with a state model

RQ4 · When could a trader have known about the state effect?

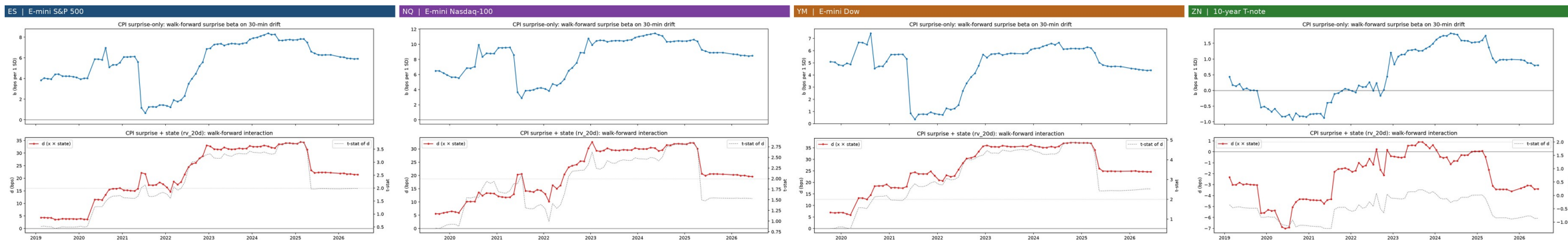


Figure 17 · Walk-forward CPI surprise beta (top) and CPI × 20-day-vol interaction with its t-statistic (bottom): ES, NQ, YM, ZN

ES

- Beta 4 bps (2019) → ~1 bp (mid-2021) → ~8 bps (2023–24)
- Interaction t crosses 2 only in mid-2022, ~3.5 in 2023–25

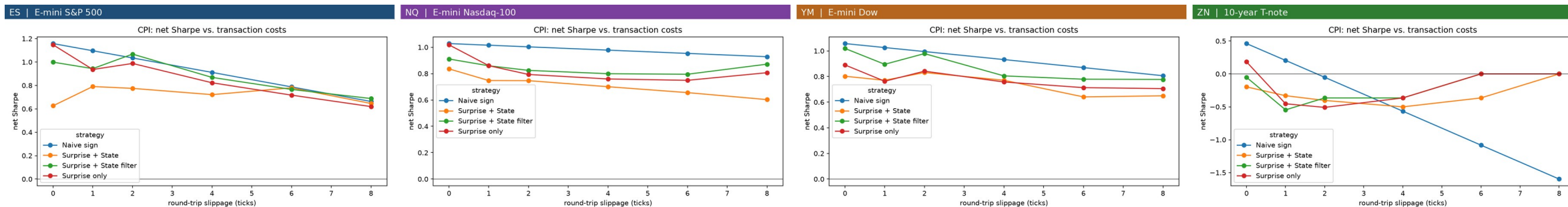
NQ

- Same shape at a higher level: 6 → 3 → 10–11 bps
- Interaction t peaks ~2.8 in 2024–25, falls to 1.5. YM: same shape, t 0.6 (2019) → 4–5 (2022–24) → 2.5

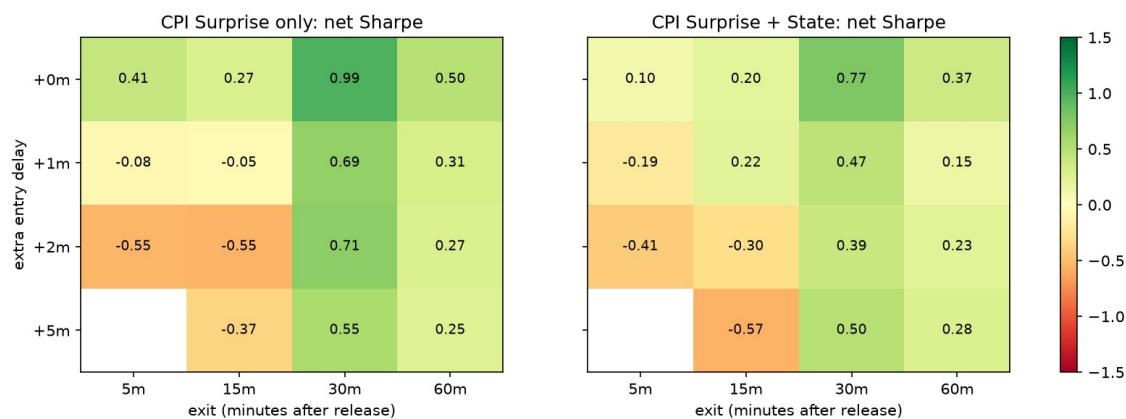
ZN

- Beta between -0.9 and +1.8 bps for the whole period
- Interaction never significant (t between -1.2 and +0.2)

RQ4 · Robust to costs and delays in equities



CPI net Sharpe vs round-trip slippage (0–8 ticks): ES, NQ, YM, ZN



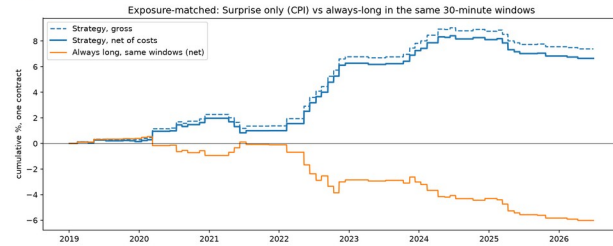
ES: Sharpe by entry delay (rows) and holding period (columns)

Robustness

- ES: 1.15 → 0.62 from 0 to 8 ticks
- NQ: almost flat (0.75–1.02); YM: 0.89 → 0.71
- ZN: turns negative at ~2 ticks
- 30-min hold survives 1–5 min delays; 5-min holds collapse
- A slow 30-minute continuation, not a speed race

RQ4 · The direction is what earns the money

ES | E-mini S&P 500



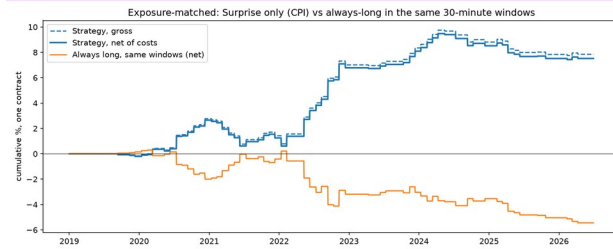
+6.6% vs -6.0%

ES: CPI surprise strategy vs always long in the same 30-minute windows

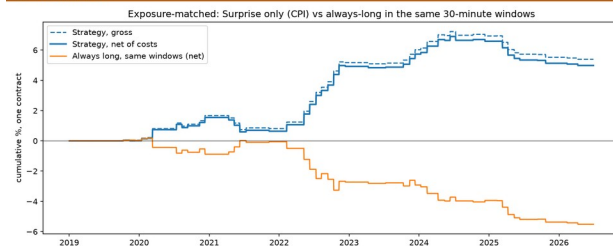
+7.5% vs -5.4%

NQ: same comparison · YM: +5.0% vs -5.5%

NQ | E-mini Nasdaq-100



YM | E-mini Dow



ZN | 10-year T-note

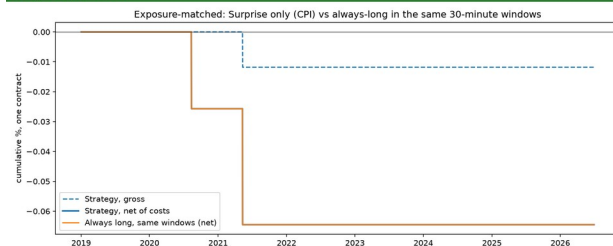
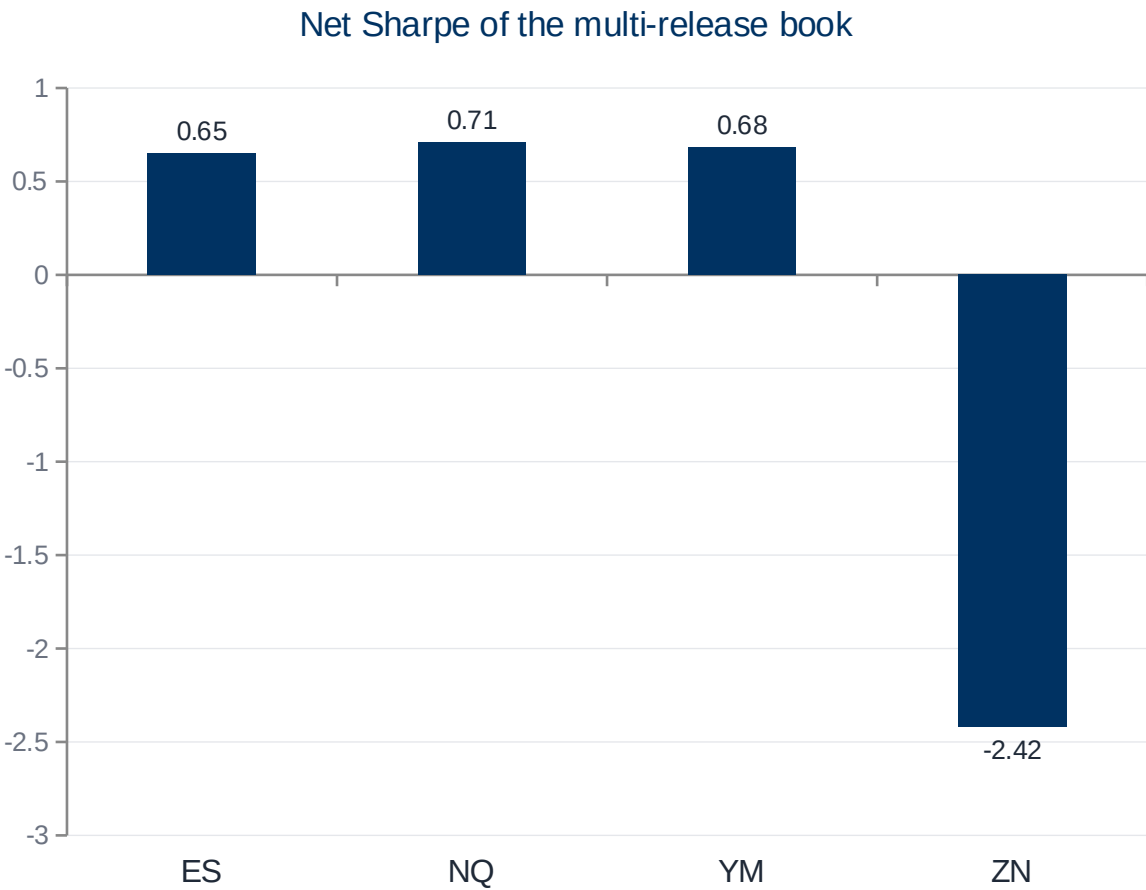


Figure 22 · the peer (DTW team) chart format

- Solid blue: strategy net of costs; dashed: gross; orange: always long in the same windows
- Being in the market at CPI time earns nothing by itself; the profit comes from the sign of the surprise (≈ 13 percentage points in ES and NQ, 10.5 in YM)
- Gross and net lines are close: costs are not the binding constraint for CPI
- Beta to buy-and-hold ≈ 0 : the strategy is market-neutral (0.06–0.1% time in market)

RQ5 · Ten releases give ~100 trades a year per market

Notebook 05: top-10 families by an impact screen (history only), naive sign, one 30-min trade per release



All · naive	ES	NQ	YM	ZN
Trades / year	98	97	93	82
Net bps / trade	1.7	2.4	1.7	-3.4
Total net P&L	12.8%	17.5%	12.0%	-20.5%
Sharpe ex-2022	0.28	0.44	0.40	-2.53
Random-sign p	<0.001	0.005	0.006	0.81
Sharpe at 4 ticks	0.20	0.60	0.45	-4.24
2nd window alone	-0.79	-0.23	-0.15	-3.08

Tenfold more trades with the edge intact in equities; the edge is gone within 30 minutes.

RQ5 · The multi-release books over time

Figure 23 · ES

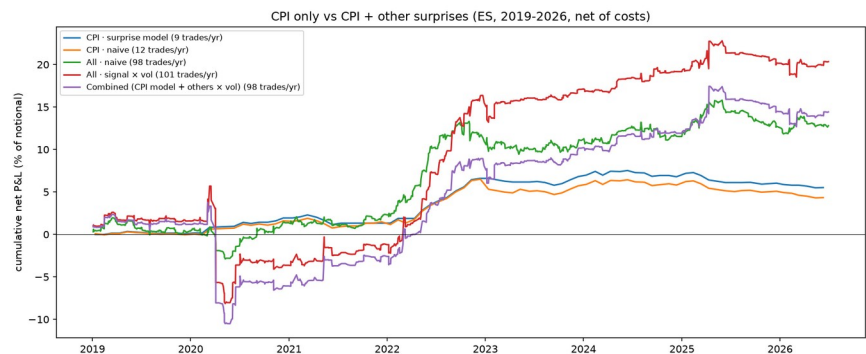


Figure 24 · NQ

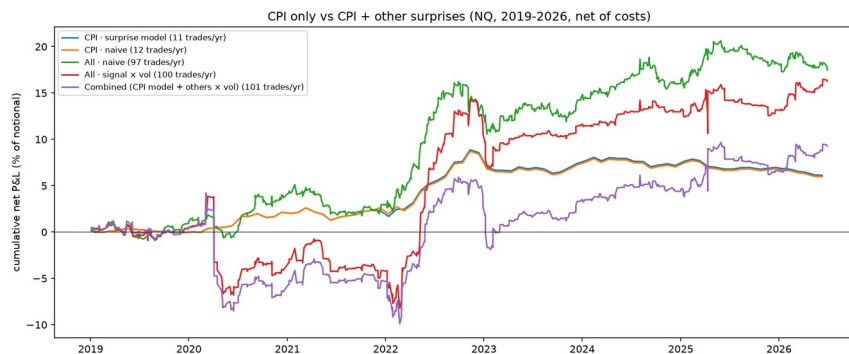


Figure 25 · ZN

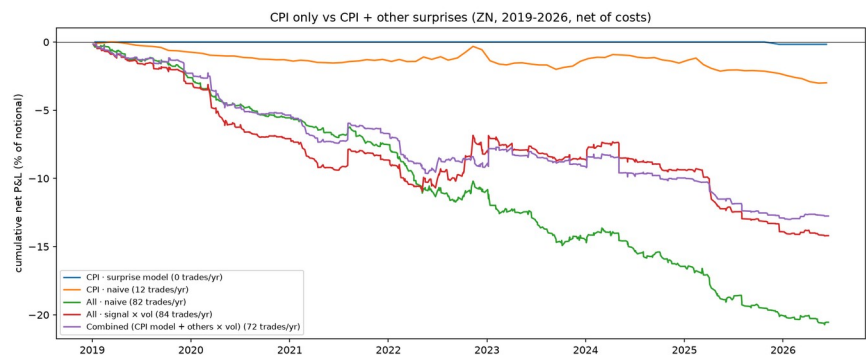
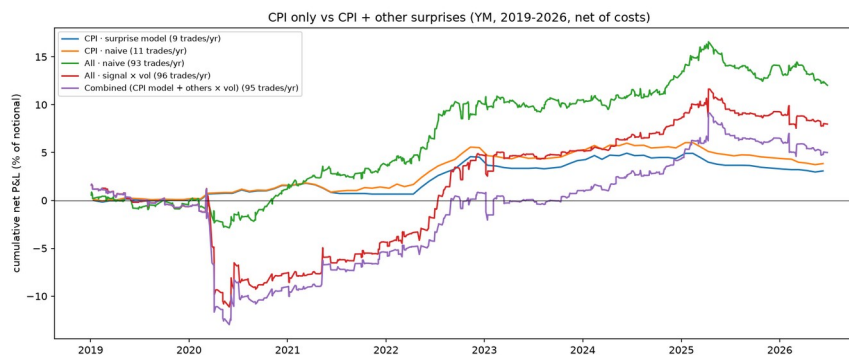


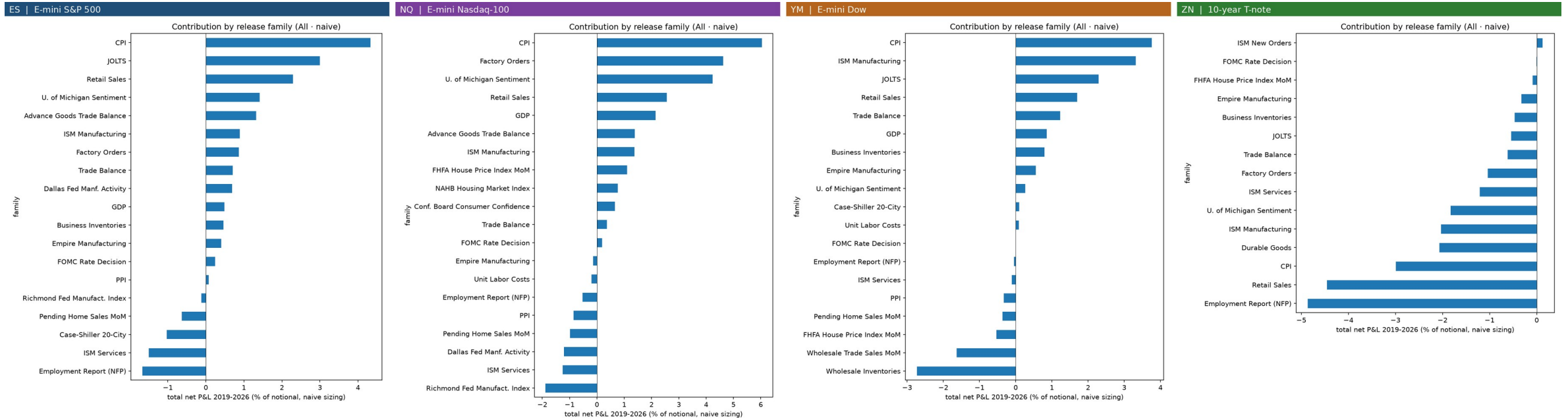
Figure 24b · YM



Reading

- ES and NQ: the naive all-release book climbs steadily, biggest gains in 2022 but positive outside it (NQ +17.5%, ex-2022 Sharpe 0.44; ES +12.8%, 0.28)
- Volatility sizing adds P&L but raises drawdowns to 12–14%
- YM: +12.0%, ex-2022 Sharpe 0.40. ZN: every book falls almost linearly, paying the cost on each trade

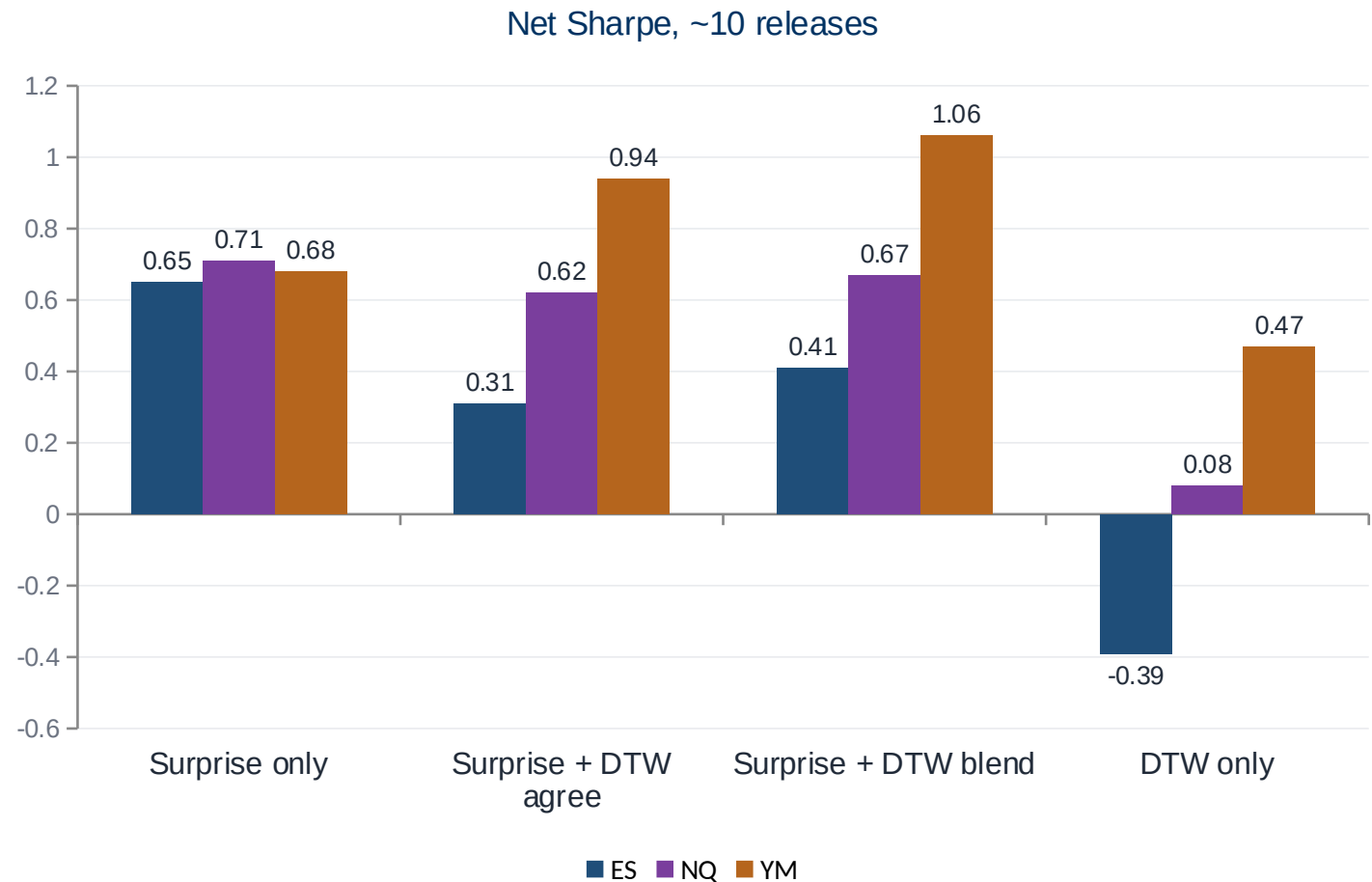
RQ5 · CPI leads; the rest is diversification across small edges



ES: CPI +4.3%, JOLTS +3.0%, Retail Sales +2.3% · NQ: CPI +6.0%, Factory Orders +4.6%, U. Michigan +4.2% · YM: CPI +3.8%, ISM Manufacturing +3.3%, JOLTS +2.3% · ES and NQ lose on the Employment Report and ISM Services · ZN: every major family loses about the cost on each trade

RQ6 · DTW adds no reliable directional information

Notebook 06: the team's DTW specification on the same releases, entry and costs



0.01 · 0.03 · 0.03 · 0.04

DTW rank IC with the 30-min return (ES · NQ · YM · ZN);
useful signals show 0.05–0.10

Takeaways

- ES and NQ: every combination lowers the surprise book's Sharpe
- CPI-only results point in different directions in ES and NQ: noise
- YM: the blend reaches 1.06 (P(not better) 0.06), not confirmed elsewhere

RQ6 · Surprise vs DTW over time

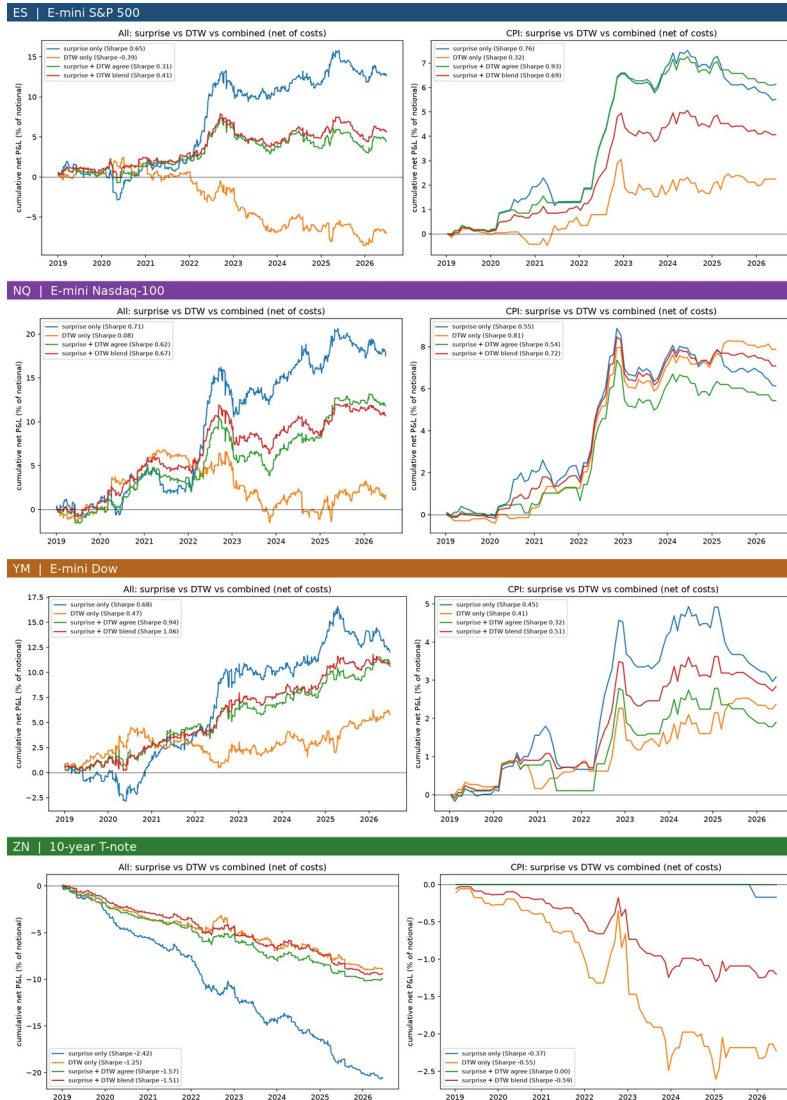
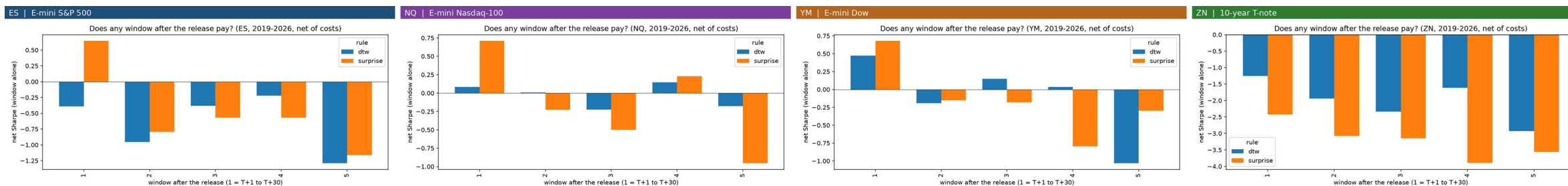


Figure 27 · ~10 releases (left panels) and CPI (right panels)

- ES: surprise only rises in steps (mostly 2022); DTW only is flat until 2022, then falls to about -7%
- NQ: DTW only rises in 2019-2022, then gives it all back from 2023, a pattern that does not persist
- YM is the exception: agree and blend beat surprise only (0.94 and 1.06 vs 0.68)
- CPI: the four lines move together; the differences are small and point in different directions in ES and NQ
- ZN: every line falls almost linearly (the cost); the surprise line falls fastest because it trades most

RQ6 · More windows add trades, not profit



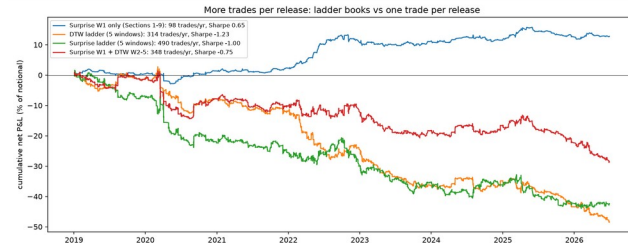
Trades/yr, Sharpe	ES	NQ	YM
Surprise, window 1 only	98, +0.65	97, +0.71	93, +0.68
Surprise W1 + DTW W2-5	348, -0.75	342, +0.27	329, 0.00
Surprise ladder (5 windows)	490, -1.00	484, -0.29	463, -0.34
DTW ladder (5 windows)	314, -1.23	309, -0.07	298, -0.23
Peer DTW slide (ES)	185, +0.61	—	—

Five windows after each release (peer structure)

- Surprise rank IC: 0.14 in window 1 (ES), ≈ 0 in windows 2-5
- The ceiling is ~ 100 trades a year per market at one trade per release
- The right way to trade more is more markets with a drift, not more windows

RQ6 · Ladder books: turnover and cost, not profit

ES | E-mini S&P 500



+12.8%

ES window 1 only: the only ES book that ends positive

-29% to -48%

ES ladder books over 2019-2026

NQ | E-mini Nasdaq-100



YM | E-mini Dow



ZN | 10-year T-note

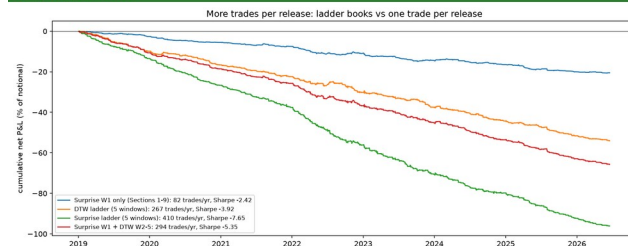
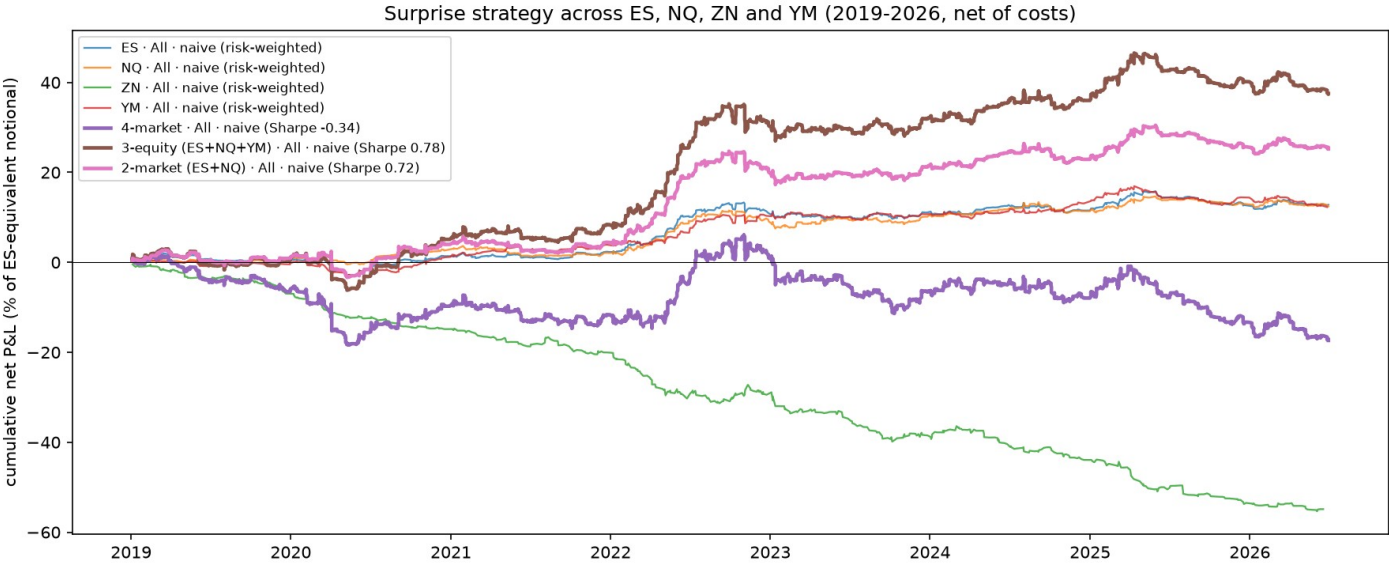


Figure 29 · three ladder books vs window 1 only

- ES: each extra window adds about 1-2 bps of cost with no edge
- NQ: cost below 1 bp, so 'surprise W1 + DTW W2-5' ends +12.9%, still below window 1 alone (+17.5%) with a drawdown three times larger
- YM: window 1 alone +12.0%, ladder books -14% to 0%. ZN: the ladder books lose 54% to 96% of notional
- More windows per release add turnover and cost, not profit

RQ7 · ES + NQ + YM combine well; ZN destroys the portfolio

Notebooks 07 and 07b: equal-risk weights from history only (ES 1.00, NQ 0.71, YM 1.02, ZN 2.67)



288

trades a year, ES + NQ + YM (195 net Sharpe (1.10 at ¼ tick); ES + for ES + NQ)

0.78

+38%

total net P&L, 2019–26

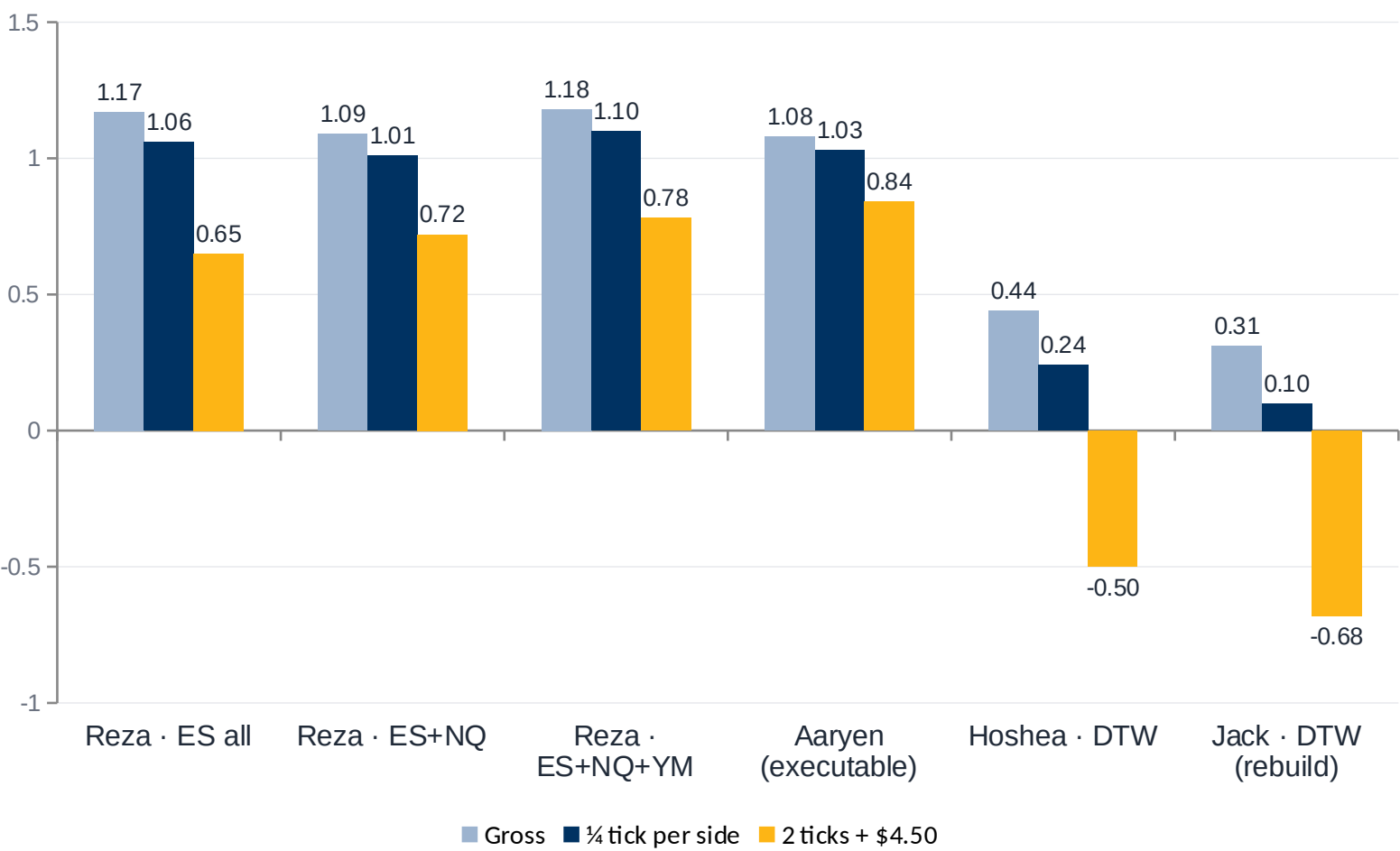
-0.34

Sharpe with ZN added (4 markets)

YM diversifies: its book correlates 0.66 with ES and 0.43 with NQ. ZN has no drift and gets the largest risk weight.

RQ7 · The team's four workstreams at equal costs

Notebooks 08 and 08b: each person's own trades re-priced on the same window (2019–2026) and Sharpe method



Takeaways

- 300+ trade-a-year DTW books are positive only at 1/4 tick
- Surprise-based books survive realistic costs
- Reza ES+NQ+YM: highest Sharpe at 1/4 tick (1.10) and highest total return at every cost
- Jack's rebuild (310 tr/yr, 0.16) is an approximation, not his slide

RQ7 · Surprise vs the team's DTW book, on identical terms

combined/surprise_dtw_book.py: Jack Duncan's data, test years 2021–2026, costs and metrics · Sharpe at $\frac{1}{4}$ tick per side

Same terms

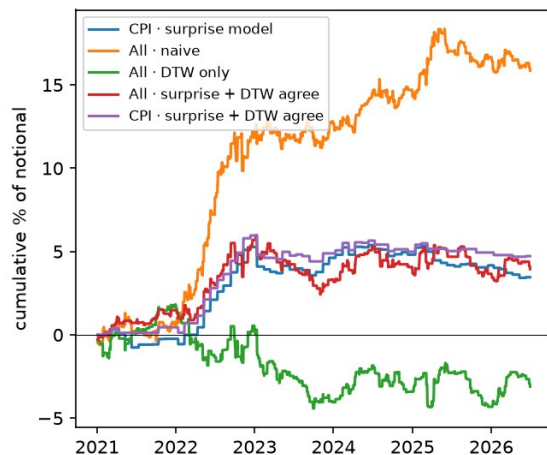
- Shared: Jack's cleaned bars, contract specs, session grid, buy-and-hold, metrics and equal-risk combined book
- Test years 2021–2026; validation 2018–2020
- Different by design: one 30-min window at T+1 (vs a five-window ladder); walk-forward refit each year from 2013

Book	ES	NQ	ZN	All 3	ES+NQ	ES trades
DTW book (Jack, 5 windows)	0.21	0.41	-0.85	-0.19	–	1,443
Surprise, top-10 releases	1.09	0.75	-0.16	0.84	1.00	527
CPI surprise model	0.51	0.46	0.17	0.47	0.49	62
Surprise + DTW agree	0.37	0.48	0.12	0.45	0.48	271
CPI + DTW agree	0.74	0.58	0.48	0.72	0.68	44
DTW only (Reza, 1 window)	-0.26	-0.28	0.19	-0.23	-0.32	356

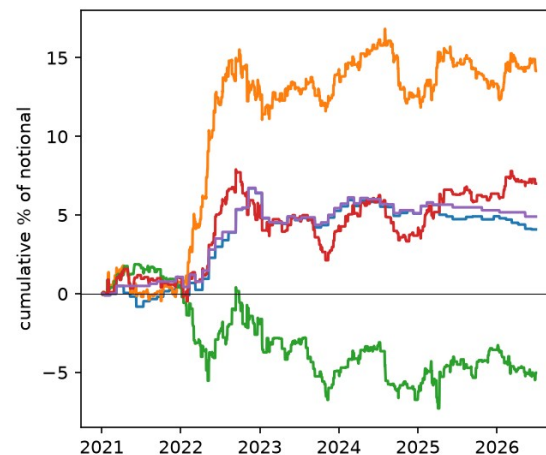
- The surprise direction beats the DTW book in both equity markets and in the combined book, with about a third of the trades
- ZN fails at any real cost in both workstreams: one tick is large next to its 30-minute moves

Adding DTW and its effect on the results

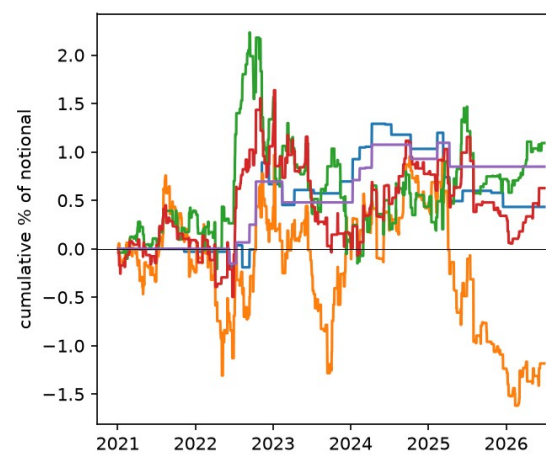
ES • E-mini S&P 500



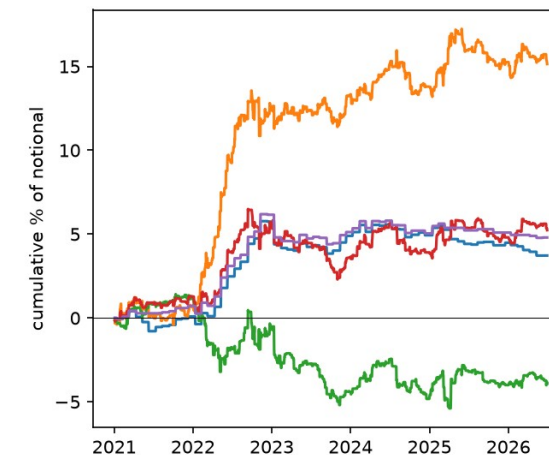
NQ • E-mini Nasdaq-100



ZN • 10-year T-note



ES + NQ • equal risk



— CPI surprise model
 — Surprise, top-10 releases
 — DTW only

— Surprise + DTW agree
 — CPI + DTW agree

Cumulative % of notional, ¼ tick per side, test years 2021–2026 (combined/surprise_dtw_book.py)

Sharpe, ¼ tick	ES 18–20	ES 21–26	NQ 18–20	NQ 21–26
Surprise, top-10 releases	1.04	1.09	0.86	0.75
DTW only	0.57	-0.26	0.78	-0.28
Surprise + DTW agree	1.46	0.37	1.21	0.48

Validation 2018–2020 vs test 2021–2026, both walk-forward out of sample.

Effect of adding DTW

- On its own (green), DTW loses in ES and NQ in 2021–2026 after working in 2018–2020: the same sign flip Jack found in dtw_dir
- As a filter on the surprise (red), it halves the trades and cuts the Sharpe from 1.09 to 0.37 (ES) and 0.75 to 0.48 (NQ)
- ZN: nothing survives costs; the surprise book (orange) ends negative
- Most of the surprise book's gain came in 2022

RQ7 · All four workstreams, three cost levels

Figure 33 · Gross (no costs)

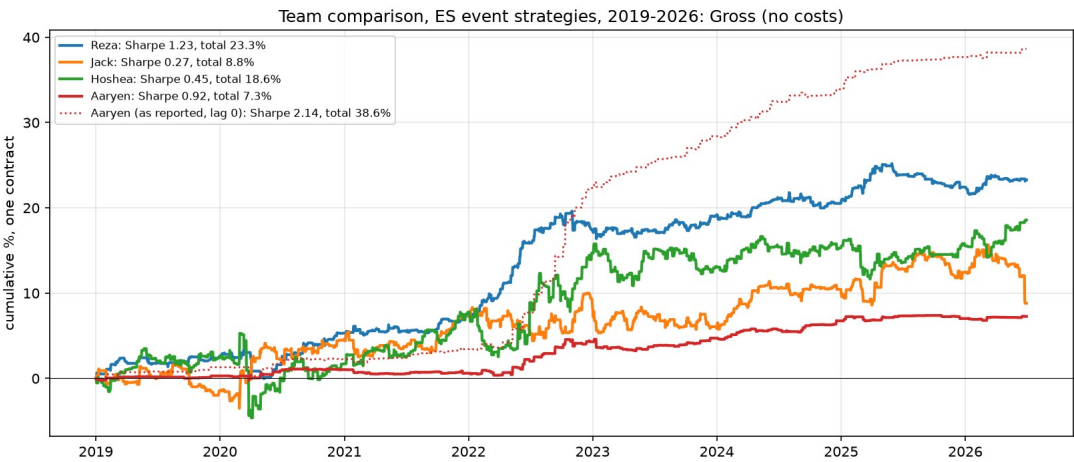


Figure 34 · Jack's cost: ¼ tick per side

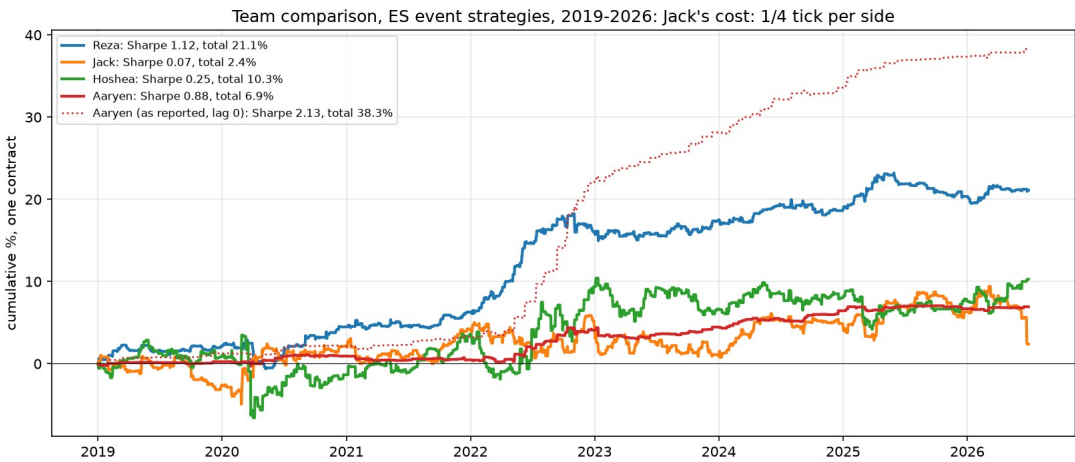
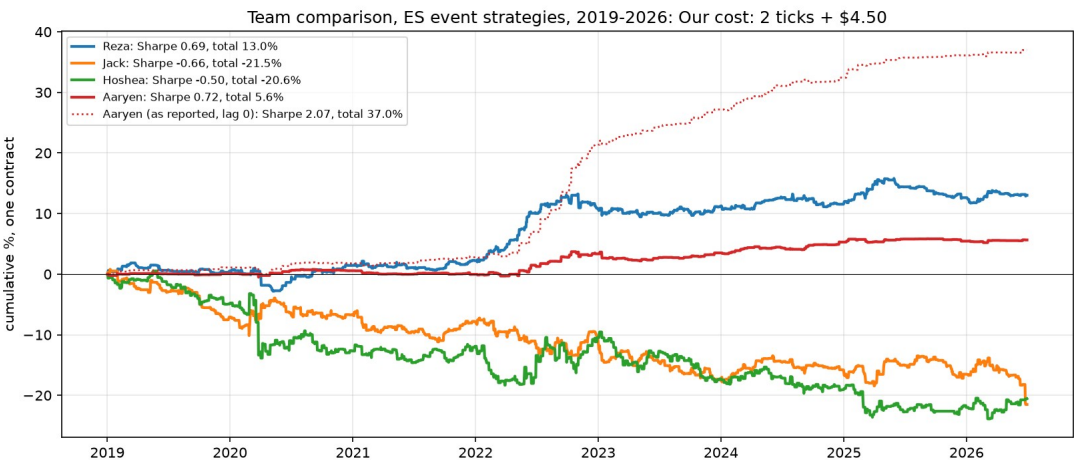


Figure 35 · Our cost: 2 ticks + \$4.50



Sharpe / total	Gross	¼ tick	2 ticks+\$4.50
Reza (ES all)	1.23 / 23.3%	1.12 / 21.1%	0.69 / 13.0%
Reza (ES+NQ+YM)	1.17 / 56.9%	1.09 / 53.2%	0.78 / 37.8%
Jack (rebuild)	0.27 / 8.8%	0.07 / 2.4%	-0.66 / -21.5%
Hoshea	0.45 / 18.6%	0.25 / 10.3%	-0.50 / -20.6%
Aaryen (exec.)	0.92 / 7.3%	0.88 / 6.9%	0.72 / 5.6%
Aaryen (reported)	2.14 / 38.6%	2.13 / 38.3%	2.07 / 37.0%

Only the surprise books stay positive at our cost; the DTW books fall to about -20%.

RQ7 · Each workstream in the peer slide format (1/2)

Figure 36 · Reza, ES all releases

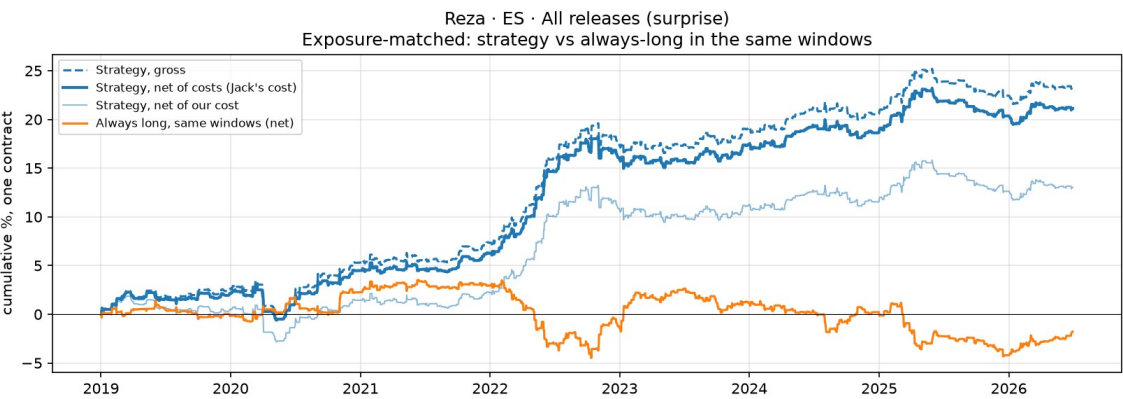


Figure 37 · Reza, ES + NQ all releases

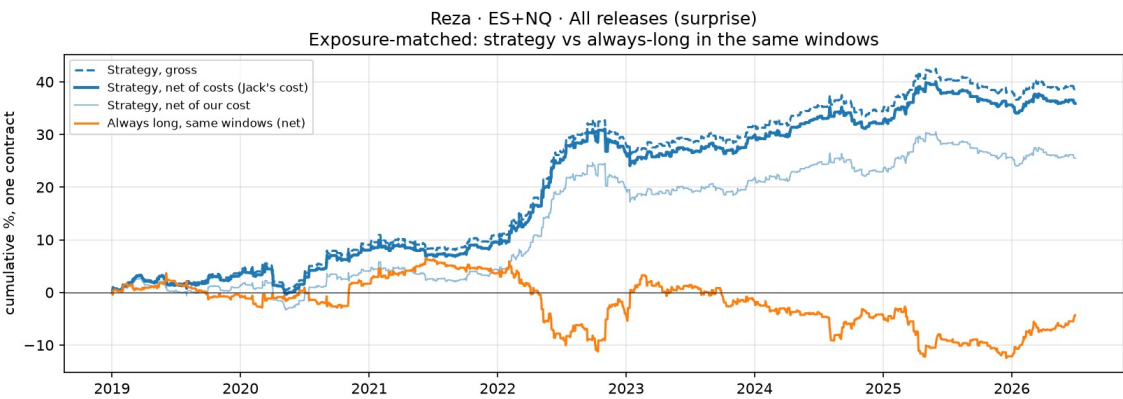
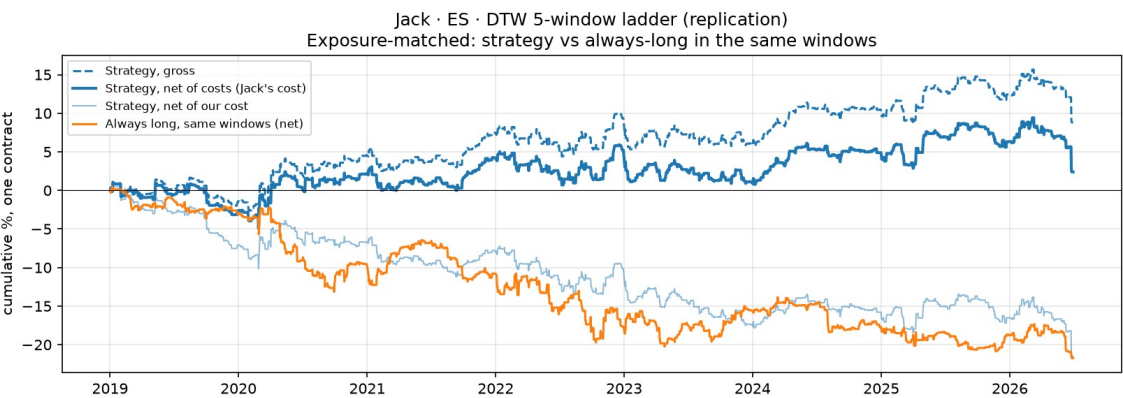


Figure 38 · Jack, DTW ladder (rebuild)



¼ tick per side	Reza ES	Reza ES+NQ	Reza ES+NQ+YM	Jack (rebuild)
Total %	21.1	35.9	53.2	2.4
Daily Sharpe	1.12	1.06	1.09	0.07
Max drawdown	-3.7%	-7.0%	-8.3%	-7.0%
Trades / yr	98	195	288	315
Always long, same windows	-1.8%	-4.3%	-9.8%	-21.7%

Jack's own slide (his data, 2018–26): 185 trades/yr, Sharpe 0.61, 18.6%. Our rebuild is an approximation.

RQ7 · Each workstream in the peer slide format (2/2)

Figure 39 · Hoshea, DTW event games

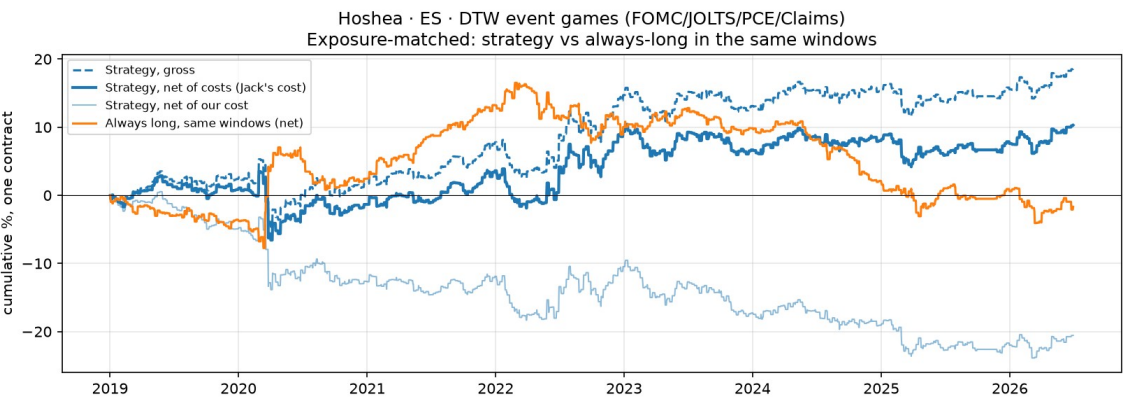


Figure 40 · Aaryen, classifier, as reported (entry at pre-release price)

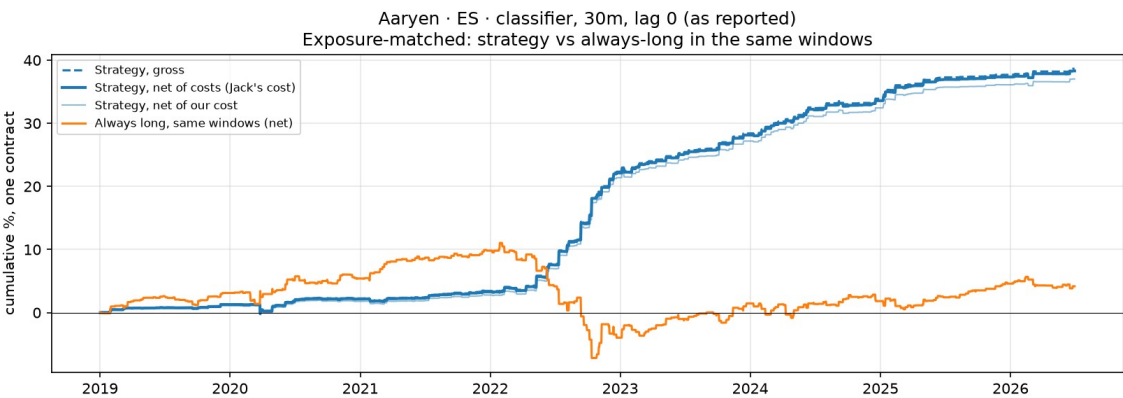
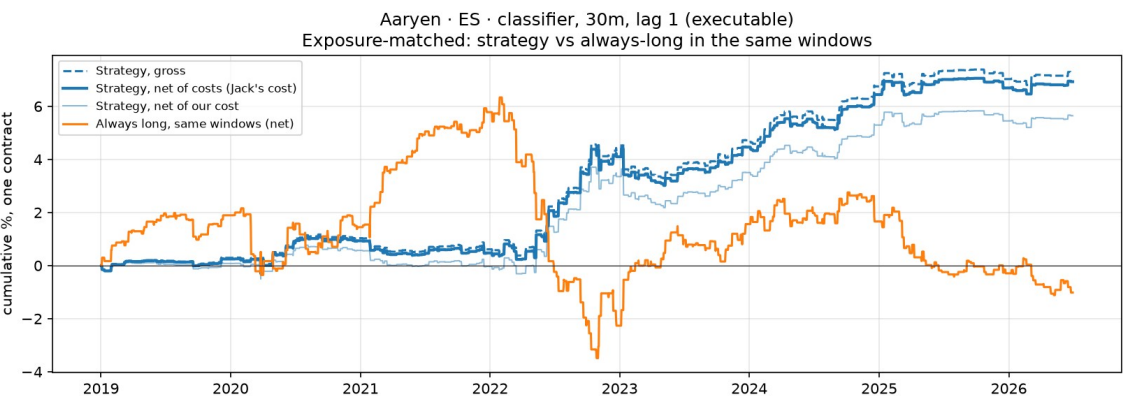


Figure 41 · Aaryen, classifier, executable (entry 1 min later)

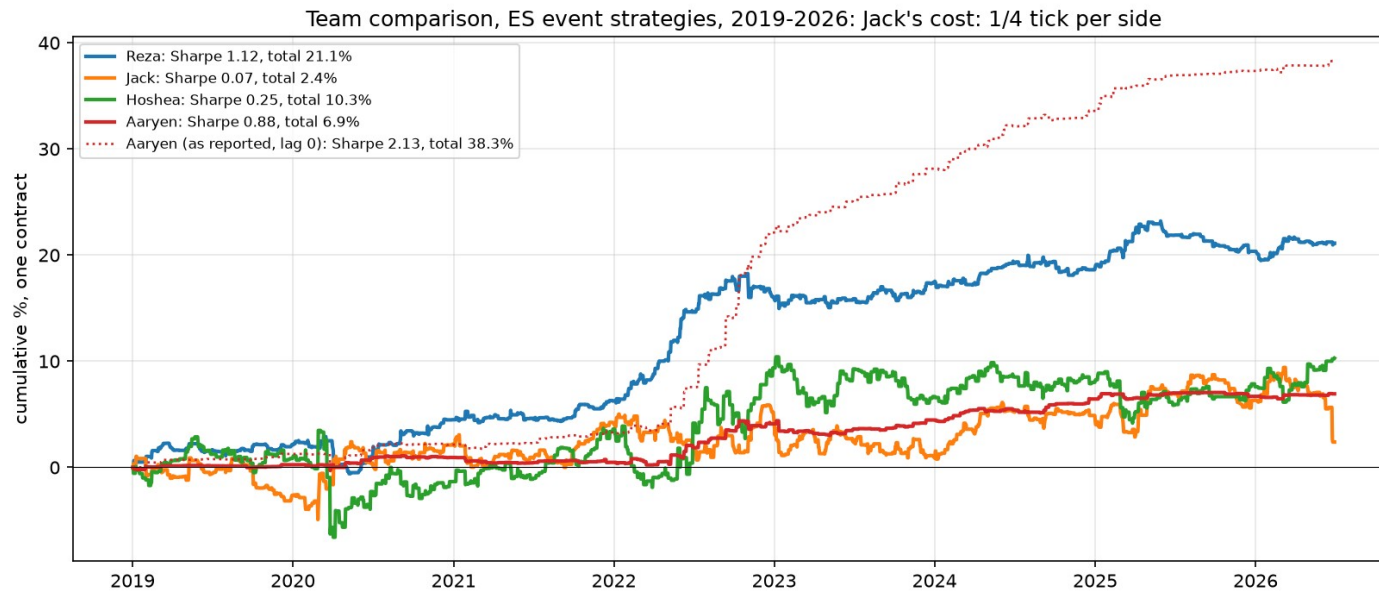


¼ tick per side	Hoshea	Aaryen reported	Aaryen exec.
Total %	10.3	38.3	6.9
Daily Sharpe	0.25	2.13	0.88
Max drawdown	-10.1%	-1.5%	-1.4%
Trades / yr	372	55	55
Always long, same windows	-1.7%	+4.2%	-1.0%

Same positions, entry one minute apart: 38.3% → 6.9%. ~80% of the reported profit is the first-minute jump.

The decisive assumption: when you enter

Aaryen's 30-minute classifier, identical positions, entry moved by one minute



2.1 → 0.9

Sharpe: entry at the pre-release price vs one minute after the release

38.6% → 7.3%

Gross total return: ~80% of the reported profit is the first-minute jump

Report every strategy with an executable entry and at two cost levels. These choices change Sharpe ratios more than any model choice.

Discussion: why equities and not bonds?

1 Interpretation takes time

A CPI print maps into yields almost mechanically; for equities investors must also judge growth, margins and risk premia. NQ (long duration) drifts most; YM behaves like a slightly smaller ES.

2 Liquidity returns gradually

Order books thin out at the release and refill over minutes: part of the drift is price discovery.

3 Bonds move first

Treasury futures price rate news directly and fastest; by T+1 the move is complete.

4 Cost relative to move

ZN: 3 bps cost vs 10 bps average CPI drift, 0.8 bps predictable. ES/NQ/YM: 1.7 / 0.7 / 1.0 bps vs 18–26 bps.

Final comparison of the three model families

Model family	Best result (net, 2019–2026)	Verdict
Surprise only	CPI ≈ 1.0 (ES, NQ, YM); 10 releases 0.65 (ES), 0.71 (NQ), 0.68 (YM); ES+NQ+YM 0.78 at 288 trades/yr	Recommended
Surprise + State	CPI 0.77 (ES), 0.75 (NQ), 0.83 (YM)	Worse out of sample; use state for risk sizing
Surprise + Pattern (DTW)	0.31–0.41 (ES), 0.62–0.67 (NQ), 0.94–1.06 (YM, n.s.)	Worse in ES and NQ; unconfirmed gain in YM
Five-window ladder	–0.75 to –1.23 (ES)	Adds only losing trades
Any ZN strategy	–0.3 to –2.4	Not tradeable at realistic costs

Only the macro surprise in the first 30 minutes after a release, in equity-index futures, gives a robust, tradeable directional edge.

Recommendations

1 Trade rule

Enter at the close of the first post-release minute, exit at T+30; sign of the family surprise; no DTW, no extra windows

2 Two-leg book

CPI core leg (surprise model or signal $\times$ vol) + multi-release naive leg on ES, NQ and YM (~290 trades/yr)

3 Risk, not direction

Scale position size by expected move (20-day vol, drawdown); stops in units of expected move

4 Clean universe

Drop families with no real surprise (FOMC, IORB); fix one CPI definition; skip $|z| < 0.5$

5 Regime scaling

Raise exposure when macro data drive policy expectations (e.g. CPI-2y yield correlation)

6 Execution & monitoring

Paper-trade 3-6 months; limit orders; rolling IC and Sharpe with pre-set stop rules

Future work

Fixed CPI definition

Pre-register one definition and a $|z|$ threshold; test on 2026+ hold-out

Order-book execution

Level-2 data to measure fills; entries at 10–60 seconds

Live paper trading

Machine-readable release feed; measure latency and slippage

More equity markets

RTY, Euro Stoxx 50, Nikkei around US releases (YM done: it works)

Short-rate markets

2-year Treasury (ZT), SOFR futures: is there a drift?

Volatility forecasting

State and DTW dispersion for sizing, stops, options

Richer surprises

Revisions, forecast dispersion, market-implied expectations

Longer sample

Back to 2000 across monetary-policy regimes

Thank you

Benjamin Steel (BlackRock) proposed the research question and supervised the project.

Ian Kaufman and Chris Pohalski (UC Berkeley Haas, MFE program) coordinated the partnership with BlackRock and the project meetings.

Jack Duncan, Hoshea Zeng and Aaryen Mehta for the DTW strategies, release screening and surprise classification credited throughout.

Code, notebooks, results, papers and the full report: github.com/rezazamani2329/BlackRock-Project-State-Dependent-Intraday-Response-of-S-P-500-Futures-to-Macroeconomic-Surprises · team code: github.com/jack-duncan/blackrock-intraday

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